LiDAR-based Wind Speed Measurement

Development of a Coherent Doppler Wind LiDAR System with Constrained Resources and Functionality



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Abstract

Characterisation of the wind field around wind turbines is an important objective in the wind-energy sector. Current methods which use anemometers mounted on met-masts are limited to single points in space, can be costly to erect, and suffer from the effects of the mast's wake on measurements. LiDAR systems mounted on wind-turbine nacelles present a remote-sensing alternative which allows for wind-speed measurement from a distance, thus providing greater spatial coverage. Doppler wind LiDAR (DWL) systems emit a beam which is backscattered by aerosols and small particles, and measure the Doppler shift of the backscatter, from which the speed of the particles, and hence the wind speed, is determined. Due to inherent challenges in measuring the small backscattered signal, these instruments are complex and require much design consideration, and for smaller research groups, scientific access is limited by the high cost of quote-only commercial systems.

In this study, an all-fibre, continuous-wave, homodyne coherent DWL system with balanced detection is designed, built and tested, with constraints on component cost and availability, particularly in the absence of an optical power amplifier. The study demonstrates the principles involved in these systems, and makes performance (primarily in terms of sensitivity) attributable to component and architecture choice. Of particular importance in the investigation is the shot-noise limit, which is the quantum limit on the system's noise, and therefore sets the best sensitivity a coherent system can achieve. The detector is shown to operate within 0.7 dB of this limit; however, phase-induced intensity noise arising from spurious reflections in the fibre paths dominates in the full system, initially raising the noise floor 22 dB above the shot-noise limit. Matching the fibre path lengths of the local oscillator and return signal reduces this to approximately 5.7 dB, though this is unstable between measurements. With this, the system is found capable of measuring wind speed at all tested flow rates (3–21 m/s) in air seeded with dense fog to account for the lower emitted optical power, with a 10.7% overshoot before calibration, attributed in part to the test rig, and an error of 1.1% after it. Recommendations are made for potential improvements of the system, particularly the addition of an optical power amplifier, a narrower-linewidth laser source, focusing optics and real-time processing.

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Chapter 1

Introduction

1.1 Background to the study

Wind has become a mainstream source of electricity. A record 165 GW of new wind capacity was installed worldwide in 2025, 40% more than the previous record year, bringing cumulative capacity to roughly 1.3 TW across 138 countries [10]. Delivering that growth at acceptable cost depends on how well the wind resource itself can be characterised, which Veers et al. identify as one of the grand challenges of wind energy science [23]. Wind speed is an important factor in analysis of a turbine's economics and engineering. The power available in the wind scales with the cube of wind speed, so a small error in measured or predicted wind speed becomes a much larger error in predicted energy [5]. Furthermore, wind speed and turbulence significantly affect the structural fatigue on a turbine, and act as an input to controllers which control the pitch and yaw of the turbines appropriately to maximise power generation and minimise stresses on the structure. It is clear that real-time measurement of the wind field around wind-turbines is a critical objective for this industry, and one to which much attention must be given. However, turbines operate in the lower atmospheric boundary layer, where the flow is turbulent and difficult to characterize [17].

The traditional instrument for wind-speed measurement for wind-turbines is the meteorological mast (met-mast), which holds cup and sonic anemometers at fixed heights, mounted on a long mast. This technology is well established, providing high resolution measurements, and indeed acting as a reference point to which any alternative method is compared [18], [13]. This methodology however holds several limitations:

Erecting a meteorological mast can be costly, particularly for offshore wind-turbines [25]. Once constructed, the met-mast is naturally limited to a single location, which limits spatial coverage - this makes variability in the wind, which occurs over larger areas, difficult to characterise [17]. Furthermore, the wake of the mast itself can corrupt measurements in the anemometer [17]. These constraints motivated the research into a remote sensing alternative to in-situ measurements.

Advancements in LiDAR technology have resulted in the technology presenting an alternative and/or complementary remote-sensing solution to existing wind-speed measurement techniques. Doppler wind LiDAR (DWL) systems measure the line-of-sight velocity of wind by emitting a laser beam into the atmosphere and measuring the Doppler shift of the beam after being backscattered by naturally occurring aerosols and particles. There are numerous commercial DWL systems available, such as Qinetiq's ZephIR, and Vaisala's (formerly Leosphere) pulsed WindCube [13] [7]. Letizia et al. identified several promising application areas for nacelle-mounted LiDAR (NML) systems, including load reduction, performance optimization, and performance assessment [17]. NML systems present several advantages over traditional measurement techniques for these ends, including larger field of view and spatial coverage (particularly for scanning LiDARs), due to its unique ability to measure wind-speed from a distance of several hundreds of meters [17].

Despite this however, scientific access to these systems is limited by high costs, with commercially available systems being quote-only, and estimated in the range of tens to hundreds of thousands of USD. Simley et al. identify system cost as one of the principal barriers to the wider adoption of LiDAR for wind-turbine applications [22]. Furthermore, commercial instruments are sold as complete, pre-calibrated products, so the design choices within them, and the effect of each choice on the system's noise and sensitivity, are not open to examination by the user. For universities and smaller research groups, this places both the instruments and an understanding of their internal trade-offs largely out of reach. It was these reasons which thus motivated this study into DWL system design within a more resource constrained environment, where there are more limitations on component availability and cost than in a commercial enterprise.

1.2 Objectives

The primary objectives of this project are to design, build, and test a Doppler Wind LiDAR system from readily available parts and components. This is done with the intention of illustrating clearly the principles involved in the design of such a system, and to make system performance in test environments attributable to design and component choices. From this a greater scientific understanding of such systems can be gained, particularly with respect to their limitations when designed and built with resource constraints. The system is not intended to form a fully functional, field-ready DWL system, with the final performance objective being limited to a benchtop-scale system, which fulfills performance requirements and illustrates the system's working principles in a constrained lab environment. The tests which are performed on the system must however illustrate a clear upgrade-path for the system, and quantify the degree of performance of the current

system, and the degree which is required for real-world use.

1.3 Scope and Limitations

The project's scope chiefly concerns the photonics, optics, and electronics-level design of a DWL system. This includes the overall choice of architecture of the system, as well as the individual components which make up the system. There are however several limitations imposed on the design of the system, and the overall scope of the project:

- **Final system performance:** As noted previously, the system is not intended, nor expected, to be usable outside of a controlled lab environment in the form presented at the conclusion of this project. However, system performance in the lab must be quantified and thoroughly investigated. For this same reason, the project is not concerned with the design of a robust, weatherproof housing and mounting for the system on a wind-turbine.
- **Component cost and availability:** The system will be limited by the cost and availability of the highly specialised components required. While a strict budget is not provided, components costs are evaluated on a case-by-case basis to determine if omission or a lower-cost option is feasible for this benchtop-scale system. The system was thus kept as simple and inexpensive as possible, given the high cost typically associated with photonics equipment.
- **Data acquisition and processing:** Real-time processing of the data is of enough complexity to warrant a separate project. Simple lab equipment is thus used for acquisition, with post-processing done on a laptop in this project.

1.4 Report outline

Chapter 2 reviews the literature on the principles of DWL systems, including the architecture types and design considerations identified by current research. Chapter 3 details the choice of architecture, the system requirements, and the design decisions made from the reviewed literature and the scope and objectives of the project. Chapter 4 gives an analysis of noise in the system, and Chapter 5 tests its functional performance in a lab environment. Finally, Chapter 6 gives conclusions based on the tested performance of the system, and recommendations for future iterations of the system.

Chapter 2

Literature Review

2.1 Aerosol Backscatter and the Doppler Shift

LiDAR systems emit very high-frequency electromagnetic radiation (when compared to its technological cousin - radar), thus, wavelengths of the emitted radiation will be very small. This property makes the technology uniquely suitable for the purposes of wind-speed measurement; the wavelength of the emitted LiDAR beam is comparable to the size of the aerosol particles, and thus the particles scatter some of the emitted beam (Mie scattering) [19].

The backscattered light can be detected by the LiDAR system's photodetector. The backscattered light, however, will be shifted in frequency relative to the incident LiDAR beam, by a magnitude which depends on the velocity of the particle itself - this is due to the Doppler effect [9]. Specifically, the shift in frequency of the backscattered light is directly proportional to the line-of-sight (LOS) velocity of the particle - that is, the component of the particle's velocity in the direction of the line of the LiDAR beam - through the following equation [17]:

$$v_{LOS} = \frac{c}{2}\frac{f_{beam} - f_{scatter}}{f_{scatter}} \tag{2.1}$$

Since the particle velocities of interest are many orders of magnitude smaller than the speed of light ($v_{LOS} \ll c$), the frequency shift is tiny compared with the optical frequency, and so $f_{scatter} \approx f_{beam}$ in the denominator. Defining the Doppler shift as $\Delta f = f_{beam} - f_{scatter}$ and using $c = \lambda f_{beam}$, where λ is the wavelength of the LiDAR beam, Equation 2.1 simplifies to

$$v_{LOS} = \frac{\lambda}{2}\Delta f \qquad \Longleftrightarrow \qquad \Delta f = \frac{2\, v_{LOS}}{\lambda}. \tag{2.2}$$

The Doppler shift is therefore directly proportional to the LOS velocity and inversely proportional to the wavelength. The factor of two arises because the light is shifted twice: once when it reaches the moving particle, and again when it is scattered back towards the

receiver. For the 1550 nm wavelength used in this work (discussed in section 3.4), each m/s of LOS velocity corresponds to a Doppler shift of approximately 1.29 MHz.

Thus, by determining the Doppler frequency shift of the received backscatter from aerosol particles, the LOS velocity of those particles can be determined, and from this, the LOS wind-speed can be inferred. Note that this explanation simplifies concepts used in ranging of the wind-speed, which will be discussed in later sections.

2.2 Challenges in Measuring Doppler Shift in DWL

A crucial challenge in DWL is the measurement of the Doppler shift itself. LiDAR beam frequencies are of the order of 10^{14} Hz (hundreds of terahertz) while, from equation 2.1, the Doppler shift in this THz-range signal, produced by aerosols moving at typical wind speeds is of the order of only 10^6–10^7 Hz (a few megahertz) - this is a fractional frequency shift of roughly one part in ten million which needs to be resolved. Photodetectors can not follow typical LiDAR optical signals, which are in the order of THz. For example, for a LiDAR system with beam wavelength 1550 nm, the field oscillates at 193 THz, orders of magnitude beyond the bandwidth of photodetectors or typical ADCs [1, 13]. Thus, the return backscatter signal's frequency can **not** be directly measured, and its frequency relative to the emitted beam simply observed from this. Instead, specialised techniques need to be used to recover the Doppler shift of the signal, **without** having to measure the frequency of the THz-range return signal for comparison. This is thus a central difficulty in DWL systems, and the detection architectures used to address it are discussed in the sections that follow.

2.3 LiDAR Detection Modes

LiDAR systems are typically grouped into two main types, each with their own distinct advantages, based on how received light is detected - these types are Direct Detection, and Coherent Detection. To varying extents, both of these detection modes have been applied in DWL systems, with significant differences in how the Doppler shift is measured in each. This section gives an overview of the principles of each detection mode, and discusses how this can be utilised to measure the Doppler shift of received backscatter light.

2.3.1 Direct Detection

Direct detection LiDAR systems operate on a very simple principle: A laser pulse is fired from the system; this pulse then gets reflected off a target (or backscattered in the case of aerosols), and the reflection is incident **directly** on the LiDAR system's photodetector. Owing to its simplicity, direct detection LiDAR systems are widely used in robotics and automotive applications for ranging and basic velocity measurements. The time between the emitted pulse and received pulse can be used to calculate the distance of targets from the system, and repeated distance measurements can be used to estimate the velocity of the target. Obtaining velocity through Doppler shift measurements however presents a challenge to this architecture.

As discussed in Section 2.2, photodetectors by themselves can not be directly used to determine the Doppler shift, as they can not follow THz frequency light signals. Multiple modified direct detection DWL system architectures have been developed to address this problem. Most common is the "double-edge technique" illustrated by Korb et al. [16]. Two Fabry-Pérot Interferometers (etalons) tuned symmetrically around the transmit laser frequency are used to split the backscatter signals - and thereafter the Doppler shift may be obtained from the ratio between the intensity of the two signals, as measured by photodetectors [8].

Crucially for this method, the ratio between scattering types (Rayleigh and Mie) for the aerosols, as well as temperature must be known in order to produce accurate measurements - this can vary significantly with altitude [26]. These factors, and strict requirements on etalon tuning, temperature, and laser stability present a considerable challenge in the implementation of this architecture [8]. In the double-edge technique any drift of the laser relative to the etalon edge is indistinguishable from a Doppler shift, so the transmit laser must be absolutely stable to roughly 100kHz of its nominal frequency, which in practice requires an active lock to an atomic transition [8]. The etalon itself must be temperature controlled to about 1 milli-Kelvin, since a 1 milli-Kelvin drift produces a $0.95m/s$ error, and mechanical drift of the mount can produce biases of $1.5 - 2m/s$ per week, requiring weekly recalibration [8].

An advantage that direct detection architectures hold over coherent detection is the ability to use molecular backscattering return (Rayleigh scattering), which gives it better performance in high-altitude regions with less aerosols to give particulate backscattering (Mie scattering, which a coherent detection system relies on) [8]. However, direct detection DWL systems typically have worse accuracy and spatial resolution than their coherent detection counterparts, which are more technologically mature [26]. For these reasons, Direct detection architectures are not as common in the literature on DWL, as compared

to coherent detection systems, particularly in typical wind-turbine applications in the aerosol-rich boundary layer, where the primary advantage of direct detection architectures is unnecessary. The remainder of this review will thus focus on coherent detection systems, and the intricacies of their system architecture.

2.3.2 Coherent Detection

Coherent detection LiDAR varies from direct detection in that the return signal (from backscatter or reflections) is optically mixed with a local oscillator (LO) signal prior to detection. Typically, a single laser source is used, where a set fraction of the beam is emitted at the target, and the other fraction is used as the LO for mixing down the line. Coherent detection provides multiple distinct advantages over direct detection, which will be described in this section.

Firstly, the optical mixing of the signals makes it possible to electronically distinguish a MHz-range Doppler shift on top of a THz-range carrier (as was identified to be a key challenge in DWL in Section 2.2). This can be described mathematically as follows: If the laser source (and therefore LO) has frequency f_0, and the return signal has frequency $(f_0 + \Delta f)$, where Δf is the frequency shift of the return signal due to the Doppler effect, the signal after mixing the return signal with LO is given by:

$$E_{mixed}(t) = \sqrt{2P_{LO}}\cos(2\pi f_0 t) + \sqrt{2P_{RS}}\cos(2\pi(f_0 + \Delta f)t) \tag{2.3}$$

Where P_{LO} and P_{RS} are the time-averaged optical powers of the local oscillator and return signal respectively.

The photodetector current, I_{det}, is proportional to the intensity of the electromagnetic field of the light signal - $|E_{mixed}|^2$. When expanded, one obtains:

$$\begin{aligned} I_{det}(t) \propto & \underbrace{P_{LO} + P_{RS}}_{\text{DC}} + \underbrace{2\sqrt{P_{LO}P_{RS}}\cos(2\pi\Delta f\, t)}_{\text{beat}} \\ & + \underbrace{P_{LO}\cos(4\pi f_0 t) + P_{RS}\cos(4\pi(f_0 + \Delta f)t) + 2\sqrt{P_{LO}P_{RS}}\cos(2\pi(2f_0 + \Delta f)t)}_{\text{not followed by detector}} \end{aligned} \tag{2.4}$$

It can be seen from the above equation that the squared mixed signal has three components: one at DC, one with frequency in the order of f_0, and another with frequency at

Δf. Note again that f_0 is in the THz range, where Δf is in the MHz range. As discussed, if a MHz-range bandwidth photodetector is used to detect E_{mixed}, the detector cannot follow oscillations at frequencies in the order of f_0, so those terms produce no response and may be omitted. Thus, Equation 2.4 simplifies to the following equation, which is called the beat signal:

$$I_{det}(t) \propto [P_{LO} + P_{RS}] + 2\sqrt{P_{LO}P_{RS}}\cos(2\pi\Delta f\, t) \tag{2.5}$$

Thus, if the frequency of this beat signal is measured, the Doppler shift, Δf, is obtained. In this way it can be seen that coherent detection allows for direct measurement of the Doppler shift, and therefore the velocity of the target.

Aside from the direct measurement of the Doppler shift, the "signal beating" process described above provides several other key advantages. The beating process amplifies the return beam, which allows for shot-noise limited operation (the ideal case for such a system), and extremely high sensitivities [13] (elaboration in 2.5). This is essential in DWL, as return signals in clear atmospheric conditions are typically very low, due to low backscatter from aerosols. Furthermore, this detection process rejects signals which are not common in spatial mode and polarisation with the LO, meaning that sunlight, or other LiDAR beams will not affect the measurement [13].

2.4 Coherent Detection: Homodyne, Heterodyne and Directional Ambiguity

The following section discusses two distinct classes of coherent detection methodology, and how the choice of class affects the recovery of the wind direction from the signal.

2.4.1 Homodyne

While not explicitly stated, the previously discussed direct down-conversion of the return signal to the beat signal - via mixing with a local oscillator with the **same** frequency as the emitted signal (Equation 2.5 and preceding) - is known as **homodyne** coherent detection. Homodyne coherent detection down-converts the MHz-range Doppler shift in the THz range carrier to a MHz-range beat signal around baseband, i.e. the frequency of the signal is directly the frequency of the Doppler shift. This has significant implications:

Ignoring the DC components in Equation 2.5, we get

$$I_{det}(t) \propto \cos(2\pi\,\Delta f\,t)\,. \tag{2.6}$$

The Doppler shift (Δf) therefore appears directly at baseband. However, because the cosine is an even function,

$$\cos(-2\pi\,\Delta f\,t) = \cos(2\pi\,\Delta f\,t)\,, \tag{2.7}$$

Therefore, noting that Δf is signed based on whether the target is approaching or receding, it can be seen that approaching and receding targets of equal speed produce identical detector currents. Only the magnitude $|\Delta f_D|$ is recoverable, and the measured spectrum is symmetric about zero frequency. Note that Abari et al. [1] demonstrated a methodology for obtaining signed velocity, even with a homodyne coherent detection setup, however this setup is novel and introduces additional complexities and costs which are outside of the scope of this project.

2.4.2 Heterodyne

The alternative to this is known as **heterodyne** coherent detection. In heterodyne architectures, a slight frequency shift is imposed on the laser - crucially however this shift is only applied to one of the fractional outputs, either the local oscillator **or** the emitted laser (but not both). This means that when the same beating process described previously is applied, the MHz-range Doppler shift in the THz range carrier is down-converted to a MHz-range beat signal centered around an intermediate frequency (f_i below), rather than baseband, as with homodyne architectures. This is conventionally achieved with an acousto-optic modulator, placed either in the local oscillator arm [12] or in the transmit arm [3]. This has the advantage of allowing for the recovery of the target's direction:

Taking the offset in the local oscillator arm, such that $f_{LO} = f_0 - f_i$, the beat term becomes

$$I_{det}(t) \propto \cos(2\pi\,(f_i + \Delta f)\,t)\,. \tag{2.8}$$

Provided $f_i > |\Delta f|_{\max}$, the sign of Δf is recoverable. Approaching targets appear above f_i and receding targets below it, and the sign of the velocity is recovered.

2.5 Coherent Detection: Noise Sources and the Shot-Noise Limit

In a coherent detection DWL system, there are four primary sources of noise which need to be considered, namely electronic noise, shot noise, relative-intensity noise, and phase-induced intensity noise:

Electronic noise arises from the photodetector and the transimpedance amplifier (which converts photodetector current into a measurable voltage), and includes dark noise (signal present when there is no light incident on the detector) and thermal noise [13]. This noise is constant, independent of LO power [7], [3].

Shot noise is a fundamental noise source in coherent detection. Shot noise arises from the discrete nature of photons, which reach the photodetector at slightly different random instants [20], i.e. it is a property of light itself, and can therefore not be removed. Because the photodetector's current is generated by these received photons, the detector then produces a white, Gaussian noise current whose power spectral density is proportional to the intensity of the light incident on it [20]. In the case of a coherent detection setup, it is thus proportional to P_{LO}. The significance of shot noise follows from comparing its scaling with that of the beat signal. From Equation 2.5 the beat current has amplitude $\propto 2\sqrt{P_{LO}P_{RS}}$, so the signal power is proportional to P_{LO}. Shot noise power is also proportional to P_{LO} [6, 3]. Electronic noise power, however, as mentioned, is constant with P_{LO}. Neglecting other noise sources for the moment, the SNR is approximately [6]:

$$\text{SNR} = \frac{P_{LO}P_{RS}}{P_{EN} + P_{SN}} = \frac{P_{LO}P_{RS}}{P_{EN} + \alpha P_{LO}} \tag{2.9}$$

Where P_{EN} is electronic noise power, P_{SN} is the shot noise power, and α is a proportionality constant representing the other terms which multiply with P_{LO} to give the shot-noise power. Observing thus that P_{SN} grows proportionally to P_{LO}, it can be seen that at low LO powers the electronic noise term dominates, and the SNR will grow as P_{LO} scales (because the numerator scales with P_{LO} as well). At very large LO powers, the electronic noise power term becomes negligible relative to the shot noise, and thus the SNR simplifies to:

$$\text{SNR} \approx \frac{P_{LO}P_{RS}}{P_{SN}} = \frac{P_{LO}P_{RS}}{\alpha P_{LO}} = \frac{P_{RS}}{\alpha} \tag{2.10}$$

This is called ideal **shot-noise limited** operation, and it is the best possible operating

point for a coherent detection system - the LO has raised the signal (and shot noise) so far above the electronic noise floor that the electronic noise floor is swamped out so as to become negligible, and no further sensitivity is available from the receiver. The SNR reaches a constant peak which is proportional only to the received backscatter signal power (P_{RS}), and the constant α which governs shot-noise power [4]. In practice however, the ideal shot-limited operating point can not be fully reached, because two effects prevent P_{LO} from simply being made as large as possible: the photodetector saturates [3], and the laser's intensity noise, discussed next, grows faster than the signal.

Relative-intensity noise (RIN) is fluctuation of the source laser's relative intensity, which appears as current noise at the output of the detector [13], creating a wideband noise floor in the frequency domain. As can be derived from Equation 2.5, the useful detectable signal **power** grows with P_{LO}. However, RIN power grows with P_{LO}^2, and thus grows faster than signal power [3]. Recall from above that we desire a large P_{LO} in order to achieve shot-noise limited operation, with larger P_{LO} giving a better SNR in Equation 2.9 **when RIN is not included**, and the ideal case being an infinitely large P_{LO}. With RIN **included** however, as in:

$$\text{SNR} = \frac{P_{LO}P_{RS}}{P_{EN} + P_{SN} + P_{RIN}} \tag{2.11}$$

It can be seen that beyond a certain P_{LO} level RIN will overtake shot noise in the denominator, and the system's SNR will no longer be increasing with increasing P_{LO} and approaching the ideal shot-limited regime [3]. Instead, the SNR will **decrease** with an increasing P_{LO} after this crossover point, due to the P_{LO}^2 scaling of RIN [6]. It can thus be seen that RIN establishes an optimal P_{LO} which can be used, and sets a ceiling on how close the detector can get to the ideal shot-noise limited operation of Equation 2.10, before increasing RIN dominates. It is therefore clear that if RIN can be reduced through other means, a larger P_{LO} can be used, and thus the system can get closer to the ideal shot-noise limited state. Balanced detection is one means to achieve this [6], and is discussed in section 2.6.

Finally, there is phase-induced intensity noise (PIIN), which is noise which results from non-ideal linewidth effects of the laser source [6], [14], [12]. PIIN is a significant source of noise in DWL systems, and has been shown to be highly unstable, and to possibly exceed the shot-noise by up to 30dB [12]. Thus, in order to remain as close as possible to the ideal shot-noise limit, PIIN should be reduced. This is possible through the selection of an appropriate laser source, and through careful matching of signal path lengths in the DWL system [12][6]. PIIN is a complex noise source which can vary significantly in shape and magnitude depending on the laser source and the DWL system. An understanding

of laser phase noise and linewidth is required to understand it - a full discussion and mathematical breakdown on these concepts and how they relate to PIIN is provided in Appendix B.

2.6 Coherent Detection: Single-Ended vs Balanced Detection

As previously discussed, in order to determine the Doppler shift of the received backscatter signal in a coherent detection DWL system, the backscatter signal must be optically mixed with the LO signal, and this mixed signal must then be detected using a photodetector system. This section discusses the two common types of photodetection methodologies - single-ended detection, and balanced detection. Balanced detection holds two key advantages over single-ended detection, which will be discussed here.

A single-ended detection system simply uses a 2:1 (2 inputs, one output) fiber optic coupler to mix the LO reference signal with the backscatter signal, and uses a single photodetector to convert this light signal into a measurable current. Balanced detection instead makes use of a 2:2 fiber optic coupler for mixing, thus giving two outputs which will be opposite in phase (due to conservation of energy [21]). These opposite phase signals are then detected using two matched photodiodes, with the output being given by the differential of the current produced by the two. At this point it is important to note that a 2:1 coupler functions exactly the same as a 2:2 coupler, but with one of its outputs terminated - the effect of this being that effectively half of the signal is discarded. In a balanced setup the two antiphase beat signals are subtracted, so their amplitudes add, doubling the beat current amplitude (a fourfold increase in signal power) relative to a single 50:50 output [15]. Because the shot noise of the two photodiodes is **uncorrelated**, the noise power only doubles, giving a net 3dB improvement in shot-noise-limited SNR over single-ended detection with a 50:50 coupler [21]. Conversely, RIN is common to both outputs and is **cancelled** by the subtraction [2] [3]. This allows the system to better approach the ideal shot-noise limit sensitivity as discussed in section 2.5. For a mathematical description of why this is the case, see Appendix C.

2.7 Transmit and Receive Optics: Monostatic vs Bistatic

For a DWL system there are two optical configurations which are commonly used - bistatic and monostatic. This section discusses the differences between the two.

2.7.1 Bistatic

Bistatic configurations use one lens to emit the laser into the atmosphere, and another to receive the backscatter signal. One distinct advantage of a bistatic configuration, is that back reflections of the beam from the lens used for emission are not coupled into the mixed signal, and thus do not appear at the detector [6]. This serves to reduce phase induced intensity noise (discussed in subsubsection 3.4.1). However, this is at the cost of greatly increased complexity, due to additional considerations of alignment of the two lens apertures [12] [4].

2.7.2 Monostatic

A monostatic configuration uses a single lens aperture for both the emission of the laser, and the receiving of the backscatter signal. A fiber optic circulator can then be used to ensure the emitted signal and the received signal do not interfere, and are routed appropriately in the system. This has the distinct advantage of not requiring any additional lenses, and alignment considerations, and indeed has become the norm for such systems due to this reduced complexity [13].

2.8 Ranging: Pulsed versus Continuous Wave

Coherent DWL systems have two distinct methodologies which can be applied in order to determine at what distance from the system a measured wind-speed is occurring. The methodology differs depending on a key architectural choice of the system, namely whether the LiDAR beam is on for longer periods of time - known as continuous wave (CW) - or if it is pulsed. Both of these architectures hold their own advantages and disadvantages, which will be discussed in this section.

2.8.1 Pulsed LiDAR

As the name suggests, a pulsed LiDAR system emits its beam in a short pulse. Depending on the pulse duration, a given volume of air - called the probe volume - will be illuminated at an instant of time, and will create backscatter. This volume naturally moves through the air as the beam propagates radially. Thus, by using extremely precise timing on the returned backscatter, it can be determined which distance "slice" of air that backscatter

was generated by, and thus, what the wind-speed for that particular "slice" is. The wind-speed for that distance is averaged over the probe volume, thus a shorter pulse will give greater spatial accuracy due to its shorter probe volume. This technique is known as range-gating, and has been widely applied in commercial and scientific DWL systems. [7] [24] [17]. A distinct advantage of range-gating is that the probe volume remains the same at all distances, including longer ones, giving it constant range resolution for longer distances than CW systems [7] (described below). However, it must also be noted that typical pulsed systems are ineffective in ranges below tens of metres, because the pulse length is at least as long as the desired range [13].

2.8.2 Continuous-Wave LiDAR

Continuous-wave systems emit a continuous beam through the atmosphere. This simplifies the design cost and complexity of the systems [7] [13], and no precise pulse timing or receive timing is required. CW systems achieve ranging by using optics to focus the beam at a specific distance. This gives a Lorentzian spatial weighting function along the beam length, with a peak at the focus point of the beam (the point with highest intensity), thus concentrating the detection sensitivity around the focus. This means that while the beam is still Doppler-shifted by varying amounts across its entire probe volume (due to variations in wind-speed across its span), the Doppler shift from the points at the focus will be weighted higher than the points not at the focus. However even with this in mind, the effective probe volume varies **quartically** with the distance of the focal point from the source, thus **significantly** increasing the volume over which the Doppler shift (and therefore wind-speed) is averaged at longer focal distances. Therefore for longer ranges, spatial resolution severely decreases, and it is not possible to gate the measurements for any specific region of air before the focal point. For this reason, CW systems are not well-suited for long ranges of several hundred meters [13], however at short ranges they can be more effective than pulsed systems, owing to the previously mentioned limitations of pulsed systems, relating to pulse length.

Chapter 3

System Design

3.1 Introduction

This chapter details the design of a benchtop prototype Doppler wind LiDAR system, whose purpose is to make the sensitivity and overall capability of a DWL measurable and attributable to specific component and architecture choices. As such, the primary considerations will be noise-levels, sensitivity, and the overarching principles of physics governing DWL systems. The design of housing for field deployment and real-time processing pipelines are thus not considered part of the scope here. Several components typical of a field-deployable DWL system are deliberately omitted in order to reduce costs, at little functional expense for a benchtop system, though where possible the system was made extensible for future iterations.

First section 3.2 gives the chosen architecture, synthesised from the architectural considerations reviewed in chapter 2. section 3.3 gives the functional and performance requirements specific to this architecture, and section 3.4 details the components chosen to build the system, based on the chosen architecture.

3.2 System Architecture

From the reviewed literature in chapter 2, an architecture of **all-fibre, coherent detection, continuous-wave, monostatic, homodyne** DWL system with **balanced detection** was chosen, with sampling and digitisation handled by a benchtop oscilloscope, and post-processing done on a laptop computer. Table 3.1 summarises why each choice was made, and the tradeoffs accepted.

Choice	**Alternative**	**Reason for choice**	**Accepted tradeoff**
Coherent detection	Direct detection	Direct detection requires the Rayleigh/Mie scattering ratio and temperature, an absolutely stabilised laser, etalon control to 1 mK, and very precise alignment and calibration [16, 8, 26], which are substantial challenges to an undergraduate project. Coherent detection is far more common in the literature, permits shot-noise-limited operation, and rejects unwanted light not mode-matched to the LO (e.g. other LiDAR systems, or the sun) [13].	Cannot use molecular backscatter when aerosol backscatter is not available; irrelevant in an aerosol-rich, controlled lab environment.
Homodyne	Heterodyne (AOM in LO or transmit arm) [12, 3]	Places the Doppler shift at baseband with no additional components; an AOM adds needless complexity and cost to a budget-constrained project.	Only the velocity *magnitude* is measured (section 2.4); acceptable as the flow direction will always be known. An AOM can be added in a future iteration without redesign.
Balanced detection	Single-ended detection	3 dB SNR improvement in shot-noise-limited operation (section 2.6), and RIN is largely cancelled, allowing the system to better approach the shot-noise limit (Appendix C).	–
Monostatic optics	Bistatic optics	A single aperture serves both transmit and receive, separated by a fibre circulator; avoids the greatly increased alignment complexity, and cost of a second set of optics and alignment equipment [6, 12, 4].	Back-reflections from the transmit path reach the detector and raise the noise floor through PIIN (subsubsection 3.4.1).
Continuous wave	Pulsed	Needs no pulse or receive timing, which simplifies design and reduces costs, and is more effective at the short ranges (tens of cm to 1–2 m) of a benchtop system, as pulsed systems are ineffective below tens of metres [13].	Reduced spatial resolution at long ranges, as resolution is defined by focus rather than gating (section 2.8); irrelevant at benchtop range.
All-fibre, non-PM	Free-space; PM fibre	Single-mode fibre makes LO–return mode-matching automatic, removes free-space alignment complexities, and gives access to widely and cheaply available telecom components [4, 1, 3]. Non-PM fibre is used owing to cost.	LO and return polarisation is not fixed, reducing the heterodyne efficiency and the measurable Doppler peak [6].

Table 3.1: Architectural choices, with reasons and accepted tradeoffs, as synthesised from the literature in chapter 2. PM: polarisation-maintaining.

3.2.1 System Architecture Diagram

Figure 3.1 below shows a functional block-diagram of the system's decided-upon architecture. In this diagram, the signal path can clearly be seen from laser source, through the splitter (splitting into LO and transmit beam). Thereafter the transmit beam gets routed via port 1 –>2 of the circulator to the monostatic optics, which also receives the backscatter signal to be routed via port 2 –>3 of the circulator. The local oscillator and backscatter signals are mixed in the optical mixer, and then detected by a balanced photodetector, from which the final output beat signal can be sampled and processed.

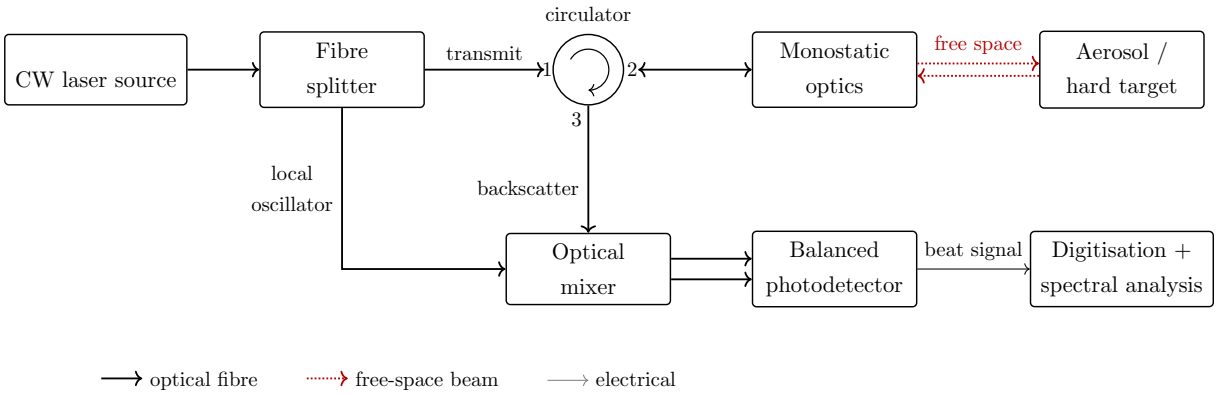


Figure 3.1: Block diagram of the chosen architecture: all-fibre, coherent, continuous-wave, monostatic, homodyne, with balanced detection

3.3 System Requirements Analysis

Now that a system architecture has been decided on, requirements which are specific to this choice of architecture, and which are related to the project goals can be synthesised. The requirements for the system are divided into two sets: functional requirements, which define the ideal outcomes which are desired from the system, and performance requirements, which define specific metrics which are to be measured and quantified with respect to the goals of the functional requirements.

3.3.1 Functional requirements

1. Measure the line-of-sight velocity of a target from the Doppler shift of coherently detected backscatter, using a monostatic all-fibre homodyne architecture with balanced detection and continuous-wave operation.

2. Operate on both a diffuse hard target of known velocity and on air seeded with

smoke or fog.

3. Produce a velocity estimate from a recorded spectrum using a stated, reproducible detection criterion (threshold relative to the system's noise floor).

3.3.2 Performance requirements

From the desired functional requirements, a set of performance requirements can be synthesised; these are given in Table 3.2 below along with the IDs of the tests which will verify them.

ID	**Requirement**	**Value**	**Verified by**
P1	Detector noise	Approaching ideal shot-noise-limited behaviour: total detector noise (electronic noise + residual RIN after balanced detection) within 1 dB of the shot-noise limit	A1, A2
P2	Phase-induced intensity noise (PIIN)	Level (in dB) and spectral shape measured; Low enough to facilitate measurements for P6	B
P3	Total system noise	Total noise floor (shot + electronic + residual RIN + PIIN) quantified with respect to the shot-noise limit, and sources of excess noise described and characterised	B
P4	Velocity detection on a hard target	Doppler peak detected at $\geq 5 \times \sigma$ (5 times the noise standard deviation) above the mean, at $N \leq 1000$ averages, for a hard target 10cm away from laser emission	C
P5	Velocity detection range	Minimum resolvable LOS speed of $\leq$ 0.1m/s, maximum resolvable LOS speed of $\geq$ 20m/s	C, D
P6	Detection of wind-speed in seeded air (smoke or fog)	Doppler peak detected at $\geq 5 \times \sigma$ (5 times the noise standard deviation) above the mean, at $N \leq 10000$ averages	D
P7	Velocity accuracy	Doppler velocity within 5% of ground-truth velocity of target (hard target and seeded flow)	C, D

Table 3.2: Performance requirements and their verification tests.

3.4 Component Choices

This section details the specific component selections which were made for the chosen architecture. These selections were made to best suit the functional and performance requirements of the system. It must be noted that this project is constrained when compared to typical scientific or industrial DWL systems. As such, many component choices were driven by cost, availability, and time-frame considerations, rather than simply their suitability for the system. This is however a suitable approach for the goals of the project, which are to illustrate the principles and physics involved in a DWL system, and to quantify system performance with respect to component choices.

3.4.1 Laser Source

The selection of an appropriate laser source is critical to ensure good performance of a DWL system. There are numerous properties of a laser source which should be considered, and which will be detailed in this section. After discussion on these properties, a suitable laser source is selected.

Wavelength

There are many suitable wavelengths for DWL systems, as detailed by Cariou et al. [6]. There is much discussion around SNR optimisation and backscatter maximisation for varying wavelengths [4], however owing largely to practical advantages, a wavelength of 1550nm has emerged as a common choice for both commercial and scientific DWL systems [1]. Fiber optics telecommunications commonly operate at this wavelength - thus making components designed for this wavelength readily commercially available and significantly cheaper than alternatives [6], [3], [13]. Additionally, laser beams at 1550nm are eye-safe at significantly higher power levels (up to multiple Watts), compared to shorter wavelengths [6], reducing safety concerns. For this reason it was decided that a 1550nm laser source would be used for this project.

Power Level

The time-averaged power of the backscatter signal into a receiver can be expressed as [13]:

$$P_s = \pi P_t \beta(\pi) \lambda \tag{3.1}$$

Where P_s is the time averaged backscatter signal power, $\beta(\pi)$ is the atmospheric scattering coefficient (evaluated at π, i.e. scattering at π radians, back towards the source), and λ is the wavelength of the transmitted light.

It can be seen in Equation 3.1 that there is a linear dependence on P_t, the power of the transmitted beam. Thus, a higher-powered transmit laser source would give a larger return signal to be measured, increasing the sensitivity of the system by increasing the SNR of the system [4].

A setup commonly used to achieve higher emitted laser power here is the master-oscillator power-amplifier (MOPA) configuration [6], [4]. This separates the functions of the laser source - allowing for a low-power, highly coherent, narrow linewidth laser source to be used as a master-oscillator and an amplifier to increase the power levels of the master oscillator beam, while maintaining its spectral purity. This also allows the local oscillator to be derived from the master oscillator, eliminating the need to have two separate laser sources - one for LO, one for output - which would need to have a stable frequency relationship [6], something which requires careful design to achieve, as illustrated by Colberg et al. [8].

As noted by Bennett, such power amplifiers (commonly Erbium-doped fiber amplifiers; EDFAs) make up a significant portion of the cost of a DWL system. Indeed, more than 20 years after the writing of Bennett's paper, the cost of EDFAs remain prohibitively expensive for a project at this scale. For EDFA systems which are able to amplify signals to hundreds of milli-Watts or multiple Watts (which is a typical power level for such a system [3][7][6]), current prices range in the thousands to tens of thousands of USD. For this reason, the use of a power amplifier for the laser source is disqualified for this project on the grounds of cost. The system was however designed such that the addition of such an amplifier would be possible at a later stage.

Due to the relatively low powers available for laser sources without power amplification (in the tens of milli-Watts), other avenues to improve detection sensitivity will be required. Bennett identifies another option for increasing the sensitivity of a DWL system - which is to average over many returns [4]. This is the approach which will be taken for this project. Additionally, owing to this constraint in laser source power, all tests which concern wind-speed measurement in air will be performed using air seeded with smoke and/or fog. This will increase backscatter levels by increasing $\beta(\pi)$ in Equation 3.1, and thus reduces the need for the power amplifier for this benchtop-scale system. The level of seeding required for a lower-power laser source to detect wind-speed is one of the investigations of this project.

Linewidth

Linewidth can be seen as a measure of the spectral purity of a laser source, and is one of the most critical parameters in the design of a DWL system. A narrow linewidth is highly desirable for DWL applications, as this will significantly reduce the phase-induced intensity noise (PIIN) in a DWL system, and allow the system to therefore have a better SNR and sensitivity to detect small backscatter signals. Furthermore, the linewidth of a laser source has a significant effect on how broad the returned backscatter signal's Doppler shifted peak is, particularly at longer ranges. This will result in the system having worse precision, and not being able to resolve turbulence in the air as well. For a full discussion on the mathematical and physical meaning of linewidth, as well as a treatment on how linewidth affects PIIN and the broadened peak, see Appendix B. Cariou et al. make the claim that linewidths typical of DFB diodes (0.5 - 5MHz, see Table 3.3) disqualify them for use in CW DWL systems due to their PIIN, however, Bennett shows that indeed a DFB source with a linewidth of 1.4MHz can be used for these purposes, provided careful attention is given to other design considerations, such as optical fiber path length matching [4]. The analysis of noise in the system, and the associated constraints arising from the use of a comparatively wider linewidth laser source indeed forms a key part of this project, owing to budget considerations, as will be discussed below and in chapter 4.

Laser source selection

There are many types of laser sources available on the market. In the selection of a laser source for this project, the primary driver was cost and availability, with the design goals being to maximise power level and minimise linewidth, while fixing the wavelength at 1550nm. Table 3.3 summarises some of the available laser sources (expanded information in Appendix D):

Source type	**Linewidth (typ.)**	**Power (unamplified)**	**Cost (USD)**
Fabry–Pérot laser diode	multi-mode	10 mW to 100 mW	< 100
DFB laser diode	0.5 MHz to 5 MHz	10 mW to 40 mW	100 to 1000
Planar external-cavity semiconductor	< 1 to 15 kHz	10 mW to 30 mW	low thousands
Er-doped DFB fibre laser	< 0.1 to 10 kHz	10 mW to 40 mW	tens of thousands

Table 3.3: Laser source classes at 1550 nm. Linewidths and powers are typical of commercial devices; costs are approximate as of September 2026 and are quote-only for the last two classes.

From these available laser source options, it was decided that a DFB laser diode would be used. While the wider linewidth, as compared to Planar external-cavity semiconductor diodes or Er-doped DFB fibre lasers is likely to substantially increase PIIN, measures such as fiber length matching will be investigated to reduce PIIN for lower-range, tabletop operation (see section 4.3), and these narrow linewidth sources are out of budget for this project. The DFB diode does however hold one advantage over the Er-doped fibre laser with respect to noise: its relaxation-oscillation peak lies at approximately 1 GHz, far above the frequency band in which the Doppler peaks of interest fall (around 5 - 20MHz), so its RIN is flat and low in the band of interest, whereas the fibre laser's relaxation peak near 1 MHz places its RIN directly in the frequency band of interest [6]. RIN is thus expected to be a minor contributor to the noise floor of this system, even before balanced detection cancels it. The specifications of the obtained laser diode are as follows:

Parameter	**Value**
Wavelength	1550 nm
Rated output power	20 mW
Linewidth	2 MHz
Operating current at 20mW	58 mA

Table 3.4: Key properties of the selected DFB laser diode (BFLD-1550-20SM-FA2). Full specifications are given in Appendix F.1.

3.4.2 Laser driver and temperature control

In order to keep the laser diode at a constant operating point, a constant current source must be used to drive it, and a temperature controller should be used. Temperature and current fluctuations present in the laser output as fluctuations in wavelength and output power [11], which can have a significant effect on the noise in measurements, particularly in a sensing system, such as a DWL, for which measurements of very small signals is required.

Perhaps the greater consideration however is the safety of the laser diode. The laser diode chosen forms the single biggest expense in this DWL system, and thus protecting it is of vital importance to this project. Laser diodes are extremely sensitive to transients and overvoltage events, which can easily damage them [11]. Circuitry for these ends may be custom-designed and manufactured as a PCB, however, this is a complex process, and is best-suited to systems where commercially available driver and temperature control circuitry is unsuitable. Thus, for this project it was decided that off-the-shelf circuitry would be used for these ends, so that the expensive laser diode is safeguarded against

possible custom design flaws or oversights of a first-time designer. Commercial off-the-shelf units integrate soft-start circuitry, current control, current monitoring, protection against overvoltage and overcurrent events and transients into a single package.

The Koheron DRV200S-A-200 laser driver was chosen, due to its very low noise characteristics, and the suitability of the current range and limit for the BFLD-1550-20SM-FA2 laser diode chosen above. The Koheron TEC100L-50 temperature controller was chosen from the same supplier, and has an adequate setpoint range and stability for the project. The respective specifications of the selected laser driver and temperature controller are given below:

Parameter	**Value**
Current range	0–200 mA
Current limit (adj.)	65–300 mA
Noise density (1 kHz)	270 pA/$\sqrt{\text{Hz}}$
RMS noise (10 Hz–1 MHz)	265 nA
Temperature coefficient	30 ppm/°C

Table 3.5: Key specifications of the Koheron DRV200S-A-200 laser diode driver. Full specs in Appendix F.2.

Parameter	**Value**
TEC current	±1.15 A
Setpoint range	10–40 °C
24 h stability	0.005 °C
Temperature coefficient	0.002 °C/°C
Supply voltage	5 V

Table 3.6: Key specifications of the Koheron TEC100L-50 temperature controller. Full specs in Appendix F.3.

3.4.3 Detector

Balanced detectors consist of two matched photodiodes, connected differentially to a low-noise transimpedance amplifier which converts the photodiode current into a measurable voltage. Since the beat signal is in the order of mega-Hertz, parasitic effects such as stray capacitance and inductance can cause amplifier instability, limit bandwidth, and introduce susceptibility to electromagnetic interference. Given the difficulty of achieving high bandwidth, low noise and high common-mode rejection in a custom RF design, a commercial balanced detector was selected so that project effort could be focused on the lidar system itself. The Koheron PD100B is used in this project, with key specifications given Table 3.7.

Parameter	Value
Photodiode responsivity	0.9 A/W
Max. input power (per diode)	1.5 mW
Bandwidth (−3 dB)	160 Hz–100 MHz
Transimpedance gain	39 kV/A
Input current noise density	8 pA/$\sqrt{\text{Hz}}$
CMRR (1 MHz)	35 dB

Table 3.7: Key specifications of the Koheron PD100B-AC balanced photodetector. Full specs in Appendix F.4.

The photodetector has a bandwidth of 160Hz - 100MHz, thus DC photocurrent is blocked (this is desired, seeing as the measured quantity of interest is the frequency power at MHz-range frequencies). This will also allow signals of up to 100MHz to be effectively detected. A beat frequency of 100MHz corresponds to a wind-speed of approximately 77.5m/s (279km/h), which is clearly far above any wind-speed which the system would reasonably need to detect. It can also be seen that the detector has a very low electronic noise floor of 8pA/$\sqrt{Hz}$, which can easily be swamped out by the shot-noise from the LO to achieve shot-noise limited behaviour, and the CMRR is high, allowing RIN to be very effectively cancelled.

3.4.4 Fiber Optics

Optical Mixer, Circulator

A standard 50:50 non-polarisation maintaining FC/APC connector type fiber optic coupler is to be used as the optical mixer (see Figure 3.1). This has two input signals which are mixed at a 50:50 ratio, and split into two output signals for balanced detection.

A standard 3-port non-polarisation maintaining FC/APC fiber optic circulator is obtained to allow transmission and reception of signals from the same optics.

Fiber Splitter

More care must be taken in selecting the fiber splitter, as the ratio here determines how much of the laser source power is directed towards emission, and how much towards use as a local oscillator for down-stream mixing.

As discussed in chapter 2, the LO power has a significant effect on the noise characteristics and the performance of the system. While the LO power can be indirectly varied by varying the laser source power (via current), this will also vary the transmit power. Thus, a splitter ratio should be selected which allows for a suitable LO power level which allows for shot-noise limited operation, when the laser is set to an output power level which provides sufficient transmit power.

The lower bound on the LO power is set by the requirement for shot-noise-limited operation, which occurs when the LO shot noise exceeds the electronic noise of the detector (see section 2.5). For the balanced detector, the shot noise current density is $I_{\text{sn}} = \sqrt{2qRP_{\text{LO}}}$ (derived from Kikuchi et al. [15]), where R is the photodetector's responsivity, and q is the charge of an electron, so equating this to the PD100B input noise density $I_n = 8\ \text{pA}/\sqrt{\text{Hz}}$ gives

$$P_{\text{LO,min}} = \frac{i_n^2}{2qR} = \frac{(8 \times 10^{-12})^2}{2(1.602 \times 10^{-19})(0.9)} \approx 222\ \mu\text{W}. \tag{3.2}$$

The upper bound is set by photodiode saturation. The 50:50 coupler divides the LO equally between the two photodiodes, which are each rated to a maximum of 1.5 mW, so $P_{\text{LO,max}} \approx 3$ mW.

For a splitter that directs a fraction κ of the laser power P_L to the LO, the condition $P_{\text{LO,min}} \leq \kappa P_L \leq P_{\text{LO,max}}$ defines the range of usable laser powers. With $\kappa = 0.1$, this gives 2.2 mW $\leq P_L \leq$ 30 mW, which spans the full output range of the 20 mW source. At full power, this yields an LO of 2 mW, approximately 9× (9.5 dB) above the electronic noise crossover and comfortably below saturation, while 18 mW is directed to transmission. By comparison, a 50:50 splitter would saturate the detector for any $P_L > 6$ mW, limiting the transmit power to 3 mW. A 99:1 splitter would provide only 0.2 mW of LO at full laser power, leaving the system electronic-noise-limited across its entire operating range. The 90:10 ratio was therefore selected, as it keeps the LO power within the shot-noise-limited window across the full range of laser drive currents. This also makes it possible to vary the LO power by adjusting the laser power across its full range, when later investigating the effect of LO power on the system's noise (see chapter 4).

3.4.5 Optics

A transmit/receive optic is required to emit the laser output into free space and to collect the backscattered light, which the circulator then routes to the detector. Since the system is a continuous-wave system and not pulsed (see section 2.8), any ranging must come from the focusing of the output beam. However, owing to the scope of the project, which is primarily to investigate the design considerations in a **benchtop** DWL system, with a

lower power laser which will only be used in a controlled, seeded air-flow in a wind-tunnel, ranging is not a primary point of investigation for the project.

A collimator was thus selected (Table 3.8). This produces a collimated beam of approximately 7 mm diameter, with a Rayleigh range of approximately 25 m - meaning that the diameter does not significantly change over 25m. In a small wind-tunnel which is much shorter than 25m, the beam diameter, and hence the receiver sensitivity, is effectively constant along the beam. The measured Doppler spectrum therefore represents the line-of-sight velocity averaged across the beam's probe volume, rather than at a single point. This removes the need to consider range in the constrained tunnel environment. In a future iteration, focusing optics could be used to select a measurement range, which would also concentrate the receiver sensitivity within the probe volume and improve the SNR.

Parameter	**Value**
Design wavelength	1550 nm
Focal length	37.13 mm
Numerical aperture	0.24
Output beam diameter ($1/e^2$)	≈7.0 mm
Full-angle divergence	≈0.28 mrad (0.016°)

Table 3.8: Key specifications of the Thorlabs F810APC-1550 fibre collimator. Beam diameter and divergence are theoretical values for SMF-28 fibre

3.5 Data Acquisition and Processing

Since real-time processing is out of scope (this typically requires high sampling-rate ADCs and FPGAs programmed to perform FFTs of the sampled data), data is obtained via a Keysight EDUX1002G oscilloscope at 125 MSa/s and post-processed on a laptop. The 62.5 MHz Nyquist frequency in a beat signal corresponds to approximately 48 m/s, far above any speed that will be measured.

Data is transferred to the laptop in discrete "records" of 25 000 samples (200 µs of data). The oscilloscope can only transfer approximately 8 records/second, while approximately 5000 records' worth of data is available in the same period, so only 0.2 % of the data in an acquisition period is used; FPGA-based systems can use up to 100 % [13]. The acquisition time for a measurement is therefore taken as the total time the data represents, ignoring the transfer bottleneck, as this is what determines the measurement rate in a real-world setup.

Using Python scripts, the power spectral density (PSD) of each capture is obtained with Welch's method: each record is divided into segments of N_{seg} samples with 50 % overlap, each segment is Hann-windowed to suppress spectral leakage, and the squared magnitude of each segment's FFT (a periodogram) is averaged over all segments of all records. Averaging after the FFT reduces the variance of the noise floor, so that a weak Doppler peak can be distinguished from random fluctuations, at the cost of acquiring more data over a longer period. The segment length sets the frequency resolution, $\Delta f = f_s/N_{\text{seg}}$, and, for a fixed record length, the number of averages, so there is a tradeoff between resolution and variance. For experiments A, B and C, $N_{\text{seg}} = 12\,500$ and 100 records are used, giving a resolution of $\Delta f = 10\,\text{kHz}$ (7.75 mm/s) and 300 averaged periodograms per capture. For the wind-tunnel tests in section 5.2, where the weak aerosol return calls for more averaging rather than finer resolution, $N_{\text{seg}} = 2500$ and 500 records are used, giving a worse (but still acceptable) resolution of $\Delta f = 50\,\text{kHz}$ (3.9 cm/s) and 9500 averaged periodograms.

Because the 50 % overlap makes adjacent segments partially correlated, these correspond to only roughly 245 and 7200 **independent** averages respectively. For stationary Gaussian noise the standard deviation of each averaged PSD bin is $\sigma \approx S/\sqrt{N_{\text{eff}}}$, where S is the noise PSD and N_{eff} the number of independent averages, giving $0.064\,S$ and $0.012\,S$ respectively. Thus, the absolute fluctuation of the noise scales with the noise floor's magnitude, while its relative size is set by the number of averages.

3.6 Final Design

The final design of the system using the architecture and components discussed above is given in Figure 3.2 below. A bill of materials is provided in Appendix E.

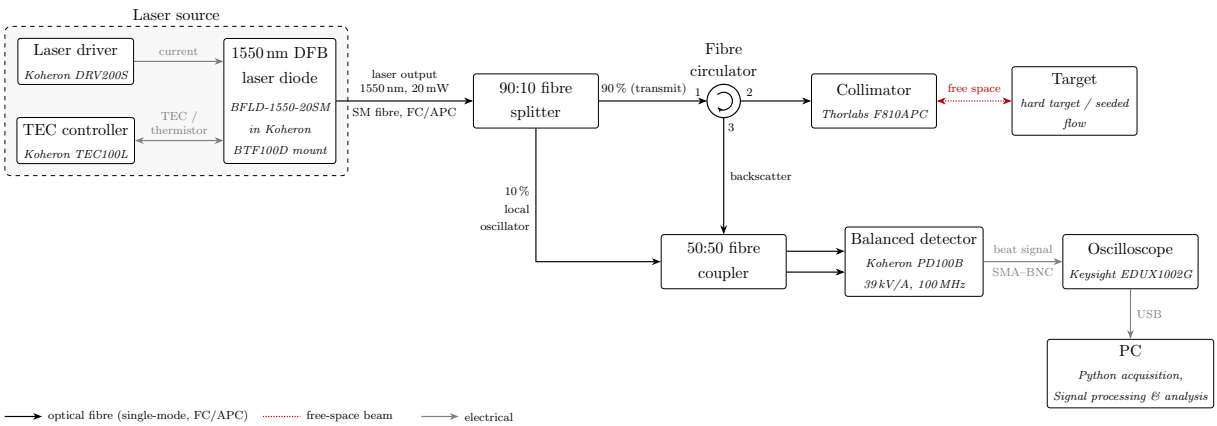


Figure 3.2: Architecture diagram of final system design

Chapter 4

Noise Analysis

4.1 Introduction

This chapter gives a comprehensive account of the testing which was done on the completed DWL system to account for the various sources of noise. The purpose of the noise analysis is to quantify the noise observed in the system and attribute it to the appropriate sources, while investigating how component, architecture, and operating point choice affects this. This chapter is separate from the functional testing of chapter 5, as the experimental setup differs significantly, and the results from these experiments will indeed be used to inform the expected results of the experiments in chapter 5, and to construct an optimal experimental setup in those cases.

4.2 A: Electronic and Shot Noise

4.2.1 Experimental Goals

As discussed in section 2.5, a coherent DWL performs best when it is shot-noise limited: the shot noise of the local oscillator (LO) must dominate every other noise source at the detector. Experiment A characterises the receiver's noise floor with the LO as the only light on the detector, so that the individual noise contributions can be separated and measured. It has the following goals:

- **Measure the electronic noise floor (A1).** With the laser off, measure the noise of the PD100B and acquisition chain alone, and compare it with the input-referred noise specified in the PD100B datasheet. Agreement with the datasheet validates the data collection and post-processing chain used in all later experiments.
- **Measure shot noise as a function of LO power (A2).** Step the laser drive current to vary the LO power reaching the balanced detector, and confirm that the

excess noise above the electronic floor grows linearly with LO power, as expected for shot noise.

- **Choose the operating point.** From the A2 results, find the LO power at which shot noise overtakes the electronic floor, and choose a laser operating point at which the system is close enough to the shot-noise limit to meet requirement P1 (Table 3.2).
- **Quantify RIN and validate balanced detection (A3).** Repeat the measurement with only one photodiode connected, making the PD100B a single-ended detector. Comparing this with the balanced case measures the laser's relative intensity noise (RIN) and confirms that the balanced detector cancels this common-mode noise as intended. This experiment is only included for completeness, to demonstrate RIN cancellation, and the results are given in Appendix G.2.

4.2.2 Experimental Methodology

The experimental setup used is given in Figure 4.1:

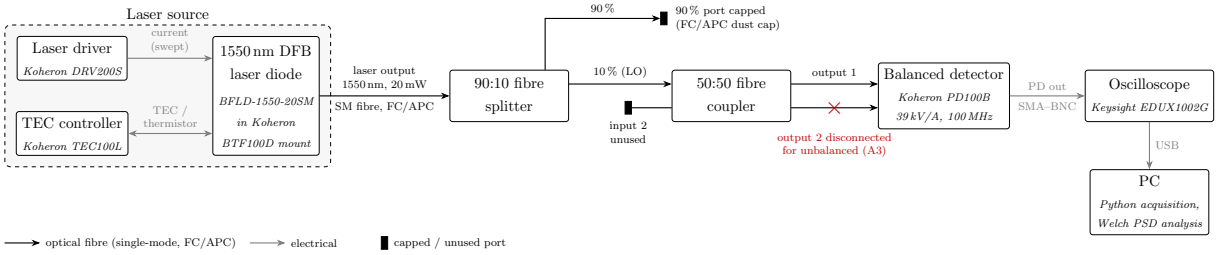


Figure 4.1: Experimental setup A

It can be seen that this differs from the full system setup. The circulator and collimator are unused for this experiment. The 90% splitter arm is left unconnected and is capped. Only the 10% splitter arm (LO) is connected to the one arm of the 50:50 coupler. Both outputs of the coupler are connected to the balanced detector, except in the case of experiment A3. Since there is no circulator, collimator or target, the only light reaching the detector is the LO itself. With no second optical field there is no beat term (as in Equation 2.5), and the LO contributes only a DC photocurrent, which is removed by the detector's AC coupling. Thus, the signal at the detector will be a pure noise signal, containing only electronic noise and shot noise. Note that phase-induced intensity noise is not present here, as it originates from reflections in the collimator and circulator which are not present in this setup, and RIN is assumed to be cancelled fully by balanced detection.

The experiment is divided into three phases, with slightly different configurations for each in order to give different measurands, as shown in Table 4.1:

ID	Configuration	Quantity Measured
A1	Laser output off, detector switched on	Electronic noise only
A2	Laser output on (current varied), detector switched on, both detector inputs connected	Electronic noise + shot noise
A3 (supplementary experiment)	Laser output on (current varied), detector switched on, only one detector input connected	Electronic noise + shot noise + full RIN

Table 4.1: Experiment A phases and configurations

For each measurement the laser is held at a fixed drive current and 100 records of the detector output are captured sequentially with the oscilloscope and transferred to a laptop. The averaged PSD of the record is estimated, as described in section 3.5. A1 requires only a single measurement, while A2 and A3 are repeated over a range of drive currents up to the 48 mA operating point.

To compare the results with a single figure, the PSD can be averaged over the 5–20 MHz band. From Equation 2.1, this band corresponds to target speeds of roughly 4–15.5 m/s, making it a reasonable band to take an average noise power measurement over, as it is where the Doppler peaks in subsequent tests would appear.

4.2.3 Results and Discussion

A1: Electronic Noise

The PSD for the measurement was obtained which gives the electronic noise from the detector only. The average PSD over the 5 - 20MHz band is calculated (using the script) as:

$$PSD_{avg(5-20MHz)} = 8.06 \times 10^{-14} V^2/Hz \tag{4.1}$$

Using the PD100B detector's gain value of 39kV/A, this can be converted into an average input noise current density across 5 - 20MHz, for comparison to the PD100B datasheet:

$$I_{input} = \frac{8.06 \times 10^{-14} V^2/Hz}{(39kV/A)^2} = 7.28pA/\sqrt{Hz} \tag{4.2}$$

Input noise current density can be seen as the equivalent of injecting this noise current

at the detector input. The calculated average of $7.28pA/\sqrt{Hz}$ across 5 - 20MHz agrees closely with the PD100B's datasheet value of $8pA/\sqrt{Hz}$, measured at 10MHz (Appendix F.4). Thus, this level of electronic noise is expected, and the measurement and post processing chain is validated.

In order to achieve shot-noise limited operation, the average PSD of the shot-noise over this region must thus be $\gg 8.06 \times 10^{-14} V^2/Hz$.

A2: Electronic Noise & Shot Noise

Figure 4.2 gives the PSD plots of the total noise - made up of shot-noise and electronic noise - for different laser currents on the same axes:

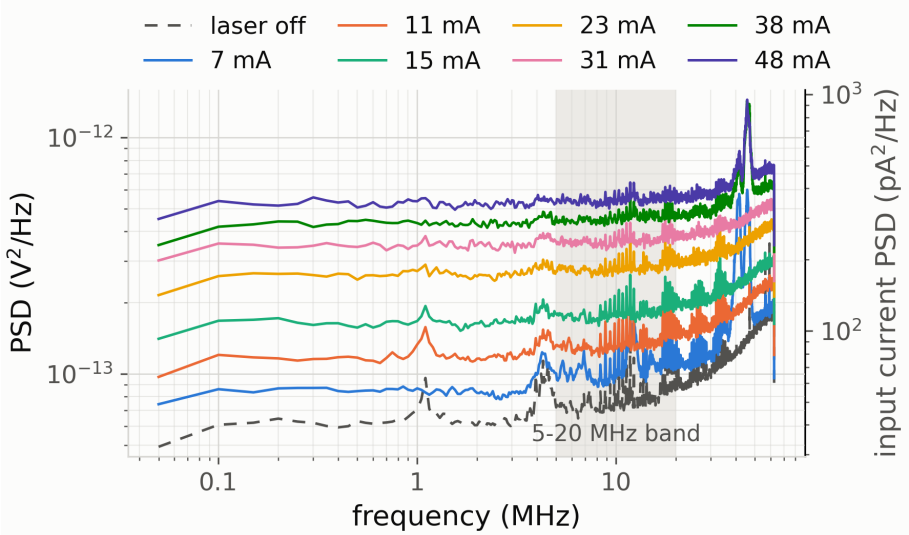


Figure 4.2: PSD of total noise (shot and electronic) for different laser source currents

It can be seen that the total noise increases with increased laser current - this is expected, due to the linear relationship between shot noise and LO power (which is related to laser current which is varied here). In order to compare the PSDs for the various laser currents, the average PSD across the 5 - 20MHz range can again be taken here, for each laser current. Figure 4.3 gives these averages plotted against laser current (table of results in Appendix G.1):

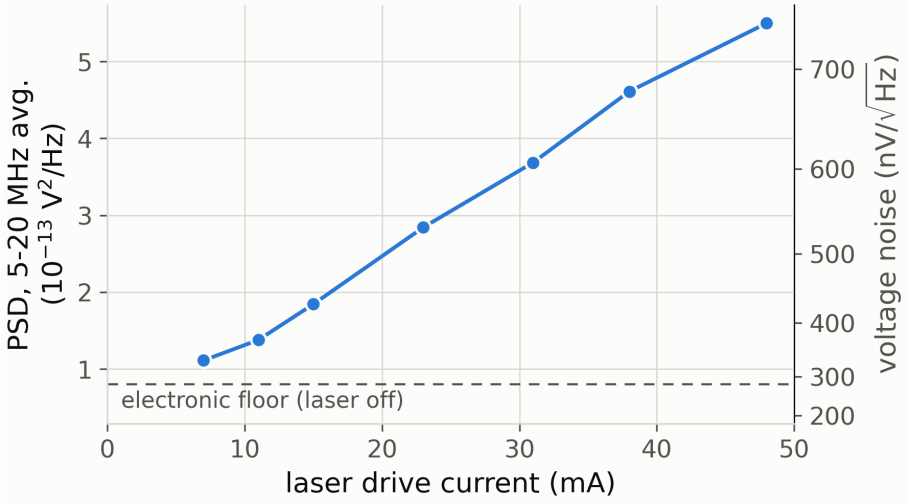


Figure 4.3: 5 - 20MHz Average PSD vs Laser Current with Balanced Detection

It can clearly be seen in Figure 4.3 and Figure 4.2 that the shot-noise exceeds the electronic noise floor at even the lowest measured laser currents, thus making the system shot-noise limited. As the current is increased, the system approaches the ideal shot-noise limit, as shot noise increasingly swamps out the electronic noise floor.

Performance requirement P1 in Table 3.2 requires that the total detector noise is within 1dB of the shot-noise limit. This requirement can be decomposed as follows:

Noise power adds in linear units, so letting S be shot noise, and E be every other noise source present here. The total noise floor, T is given by:

$$T = S + E = S(1 + \frac{E}{S}) \tag{4.3}$$

Converting to dB:

$$10log_{10}(S(1 + \frac{E}{S})) = 10log_{10}(S) + 10log_{10}(1 + \frac{E}{S}) \tag{4.4}$$

In this form, it can clearly be seen that as S increases, the effect of E on the total noise floor is increasingly reduced, as the second term, $10log_{10}(1 + \frac{E}{S})$ approaches zero. This second term is the excess noise above the ideal shot-noise limit, which is given by $10log_{10}(S)$. Thus, in order to obtain a total noise floor which is within 1dB of the shot noise limit, we set:

$$10log_{10}(1 + \frac{E}{S}) = 1dB \tag{4.5}$$

$$\rightarrow \frac{E}{S} \leq 0.26 \tag{4.6}$$

Substituting the obtained electronic noise floor obtained in A1 into Equation 4.6, we obtain:

$$S \geq 3.1 \times 10^{-13} V^2/Hz \tag{4.7}$$

Noting that Figure 4.2 and Figure 4.3 both give the **total** detector noise floor (shot **and** electronic noise), we require the total noise to be above $3.906 \times 10^{-13} V^2/Hz$ to give a shot noise value of $3.1 \times 10^{-13} V^2/Hz$. This can be seen to be reached at a laser current of approximately 32mA. Above 32mA, the total detector noise floor is within 1dB of the ideal shot-noise limit, and the ideal shot-noise limit is further approached as the current is increased.

From this, it can be seen that an operating current of 48mA for the laser allows P1 in Table 3.2 to be met with the highest margin, giving a shot noise value of $4.49 \times 10^{-13} V^2/Hz$, and a total combined noise of $5.29 \times 10^{-13} V^2/Hz$, and this will thus be used as the laser current for the remainder of this project. While a higher laser current still may be used, to even further approach the ideal shot-noise limit, this risks saturating the detector. The detector has a maximum incident power rating of 1.5mW on each photodiode (Appendix F.4). From the laser datasheet (Appendix F.1), a current of 48mA gives approximately 17mW at the laser output, which will result in approximately 0.85mW at each detector from the LO alone. This allows the detector to thus remain comfortably far away from its saturation point, even if returned signal is high and results in a higher incident power at the photodiode after mixing with the LO.

A3: Relative Intensity Noise

Experiment A3 serves only for completeness, to illustrate the effectiveness of balanced detection in removing relative intensity noise at the detector output. It was found that only 0.79 dB of excess RIN was present at the 48 mA operating point, which balanced detection then cancelled. This low level is consistent with the choice of a DFB diode (section 3.4), whose relaxation-oscillation peak lies far above the 5–20 MHz measurement band, and confirms that RIN is not a significant noise source in this system; PIIN, discussed next, is. See Appendix G.2 for the full results and discussion of this supplementary experiment.

4.3 B: Phase-induced Intensity Noise

4.3.1 Experimental Goals

As discussed in section 2.5, and elaborated in Appendix B, Phase-induced Intensity Noise (PIIN) can be a significant contributor to the total noise of the system, rising far above shot-noise - particularly when the linewidth of the laser source is not narrow, and when there is significant path-length mismatch between the LO and the transmit signal.

The goal of Experiment B is to measure and quantify the magnitude and shape of the total noise of the system, including PIIN, in a controlled setting. There is effectively no target in this experiment, thus making all PIIN in the measurement attributable to reflections from the circulator and collimator, and independent of target.

4.3.2 Experimental Methodology

For this experiment, the system is set up in full, as given by the final design in Figure 3.2. The collimator is then pointed at a stationary matte-black painted surface, angled 45 degrees away from the collimator, and at a distance of 10cm from the collimator. Any reflections from the target surface will not be Doppler shifted, because it is stationary - thus, if these reflections are sent back into the collimator, they will appear at the output as additional PIIN (see Appendix B). This setup thus intends to absorb and reflect a large amount of the emitted light **away** from the system, so that the measured PIIN can be attributed to only the internal reflections in the system from the collimator and circulator, which is independent of target.

The laser is then driven with a current of 48mA (the chosen operating point for LO power), and the same oscilloscope acquisition chain and post-processing chain (Welch PSD) as in experiment A is used to obtain the PIIN power. As discussed in [14], PIIN can vary drastically depending on temperature, or small disturbances in the system's fiber optics, thus this measurement is repeated 10 times over the course of an hour to account for such variations. The average noise power as well as the spread from these measurements can then be computed.

4.3.3 Results and Discussion

Initial Results

The total noise was found to be significantly above that measured in A2, owing to the excess noise introduced by PIIN. In the 5–20 MHz band, the total noise was measured to be 22.1 dB above the shot-noise floor. PIIN is thus the largest single noise contributor in the system, and the system is therefore no longer shot-noise limited. Experiments C and D (chapter 5) were first attempted using the system as is, with the PIIN-limited behaviour described above. While the system could adequately detect the velocity of a hard target, it did not have enough sensitivity to detect the velocity of a seeded air-flow, owing to the high noise floor from PIIN. It was thus decided that an attempt would be made to reduce PIIN, from the theory discussed in section 2.5 and elaborated in Appendix B, and the experiment would be rerun for this improved system.

Reducing PIIN: Methodology and Results

Obtaining a narrower linewidth laser source is out of budget for this project, so the only remaining option was to better match the fibre optic path length between the LO and the internal reflections which mix with the LO to cause PIIN. This allows the LO and the reflections to be more coherent in phase at the mixer, resulting in lower PIIN (see Appendix B). FC/APC fibre patch cords of varying lengths were inserted between the 10% splitter arm and the 50:50 mixer, to increase the LO path length to more closely match the longer path followed by the reflections. The total system noise was measured for each patch cable length using the same methodology, with 10 captures each. The results are given in Figure 4.4 and Table 4.2, with the electronic noise floor and the total noise floor (excluding PIIN) from Experiment A2 given as reference.

Configuration	**Mean PSD** ($10^{-12}\,V^2/Hz$)	**Total noise above shot noise (dB)**	**PIIN above shot noise (dB)**	**Measured reduction (dB)**
No matching	85.5	22.1	22.1	0.0
2 m cable	27.9	17.2	17.1	−4.9
3 m cable	18.4	15.4	15.3	−6.8
4 m cable	1.98	5.7	4.3	−17.7

Table 4.2: Total system noise in the 5 MHz to 20 MHz band for each LO-arm patch cable. The PIIN alone is the total noise with the Experiment A2 noise floor (the shot-noise reference) subtracted in linear units, and the measured reduction is the change in this PIIN-alone level.

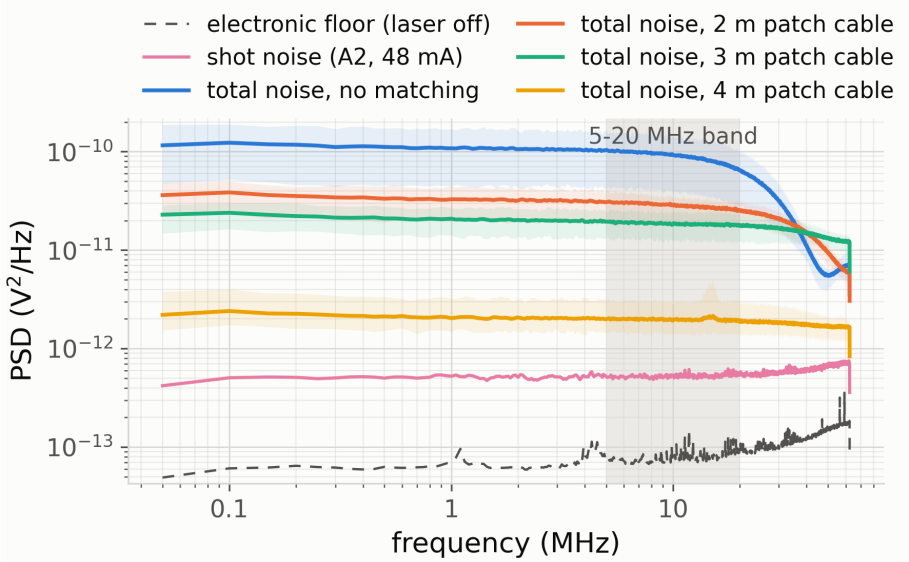


Figure 4.4: Total noise in the system for different inserted patch-cable lengths

As expected from the theory of Harris et al. and Jiang et al. [12] [14], PIIN varied significantly, even between measurements with the same setup. It can also be observed that the shape of the PIIN corresponds to the $L_c \gg \Delta L$ regime described in Appendix B, as it is a flat floor and does not contain a Lorentzian peak at baseband.

It was found that increasing the patch cable length (and thus increasing the fibre path matching) significantly reduced the total noise, with the PIIN reduction loosely matching the ΔL^2 dependence identified by Harris et al. [12]. The 4m patch cable gave a remarkable 17.7dB reduction in PIIN, bringing the total noise to just 5.7dB above shot noise. While the system is still PIIN-limited, it is far closer to the ideal shot-noise limit, and this gain in sensitivity is what made the wind-speed measurements of Experiment D possible. The 4m patch cable is therefore kept in the system for all results reported in chapter 5.

Further reduction through path matching alone is limited, since the PIIN arises from several reflections at different points in the transmit path, and a single LO patch cable can match the path length of only one of these; resulting in there still being some residual PIIN. In the $L_c \gg \Delta L$ regime observed here, the PIIN level is also however proportional to the laser linewidth Δf (Appendix B) [12]. Thus, replacing the 2 MHz DFB with a 100 kHz source (for example an external-cavity or narrow-linewidth fibre laser) would reduce the PIIN by approximately 13 dB (from approximately 2.7x the shot noise level to 0.14x the shot noise level), lowering the total noise from 5.7 dB to about 0.6 dB above the shot-noise limit, an improvement of approximately 5 dB in sensitivity. A narrower linewidth would also reduce the linewidth broadening of the Doppler peak (Appendix B), concentrating the return in fewer frequency bins and further improving its detectability.

Chapter 5

Functional Testing

This chapter details the testing which was done on the system's primary function: velocity measurement. Two distinct test setups are used, which are described in this section - results were then collected for these test setups and are presented and discussed with respect to the goals of the system and the overarching project.

5.1 C: Hard-target Speed Measurement

5.1.1 Experimental Goals

Before attempting wind-speed measurement, where the aerosol return signal is very weak, the working principle of the system is first to be validated on a hard target, whose return signal is far stronger and whose velocity can be set precisely. This isolates set-up and processing problems from the difficulties of a weak signal, and tests performance requirements P4 (hard-target SNR), P5 (low-speed detection) and P7 (velocity accuracy).

5.1.2 Experimental Methodology

A disc driven by a NEMA-17 stepper motor provides a constant, accurately known target speed. The 3D-printed rig in Figure 5.1 holds the collimator at $\alpha = 45°$ to the motor axis, aimed at a point a distance $d = 15\,\text{mm}$ from the disc centre. Since the disc is at an angle to the collimator, there will be a component of its angular velocity which is in the direction (line-of-sight) of the beam. The line-of-sight velocity at that point is

$$v_{\text{LOS}} = \frac{2\pi N}{60}\, d \sin\alpha, \tag{5.1}$$

where N is the motor speed in RPM. Two features of the geometry affect the measurement. Any error in d or α scales v_{LOS} directly. Additionally, the collimator output beam is **not** infinitesimally small, and thus the finite beam spot (which is about 7 mm in diameter)

covers a range of radii on the disc, and hence covers a number of line-of-sight velocities, which broadens the Doppler peak.

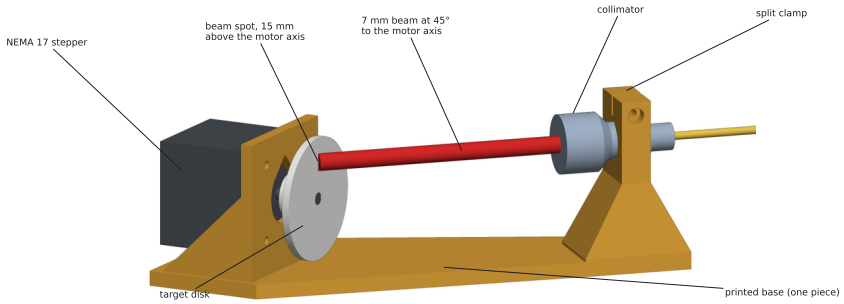


Figure 5.1: Test setup used for hard-target speed measurement

The PLA disc transmits much of the light at 1550 nm, allowing returns from stationary parts behind it, so white paper was attached to its face to provide a strong, diffuse return. The rest of the system was configured as in Figure 3.2, with the laser at 48 mA. For each motor speed, 100 oscilloscope records were captured, and their Welch PSDs averaged, using the same processing chain as experiments A and B, as detailed in section 3.5. From Equation 2.2, the expected Doppler shift is 1.29 MHz per m/s of v_{LOS}.

5.1.3 Results and Discussion

Figure 5.2 below shows the PSD of the measurements obtained for different motor RPMs tested. Table 5.1 gives the Doppler peak and local noise floor for each target speed. The Doppler peak, f_D, is taken as the power-weighted centroid of the excess PSD at each peak, as compared to the stationary-target reference, and $v_{\mathrm{LOS}} = f_D/(1.29\,\mathrm{MHz\,s/m})$. The noise statistics are taken in a band outside 3×FWHM around the centroid, out to 2 MHz either side. All PSD values are in the same units, so the PSD at f_D can be compared directly with the detection threshold (mean $+\,5\sigma$). The excess power created at the Doppler frequency is calculated by integrating each peak across the frequencies which it spans. This integrated excess is the mean-square voltage of the beat signal, which Equation C.4 relates directly to the received optical power.

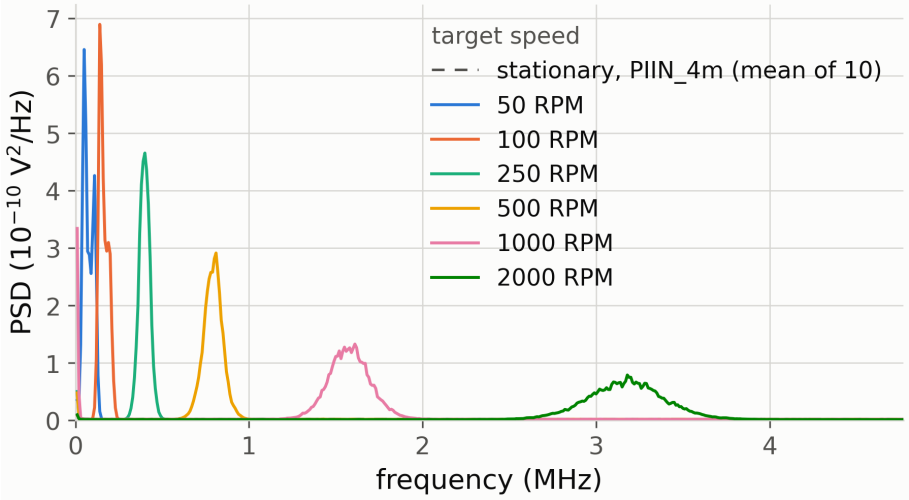


Figure 5.2: C: PSD of hard-target speed measurements, at different motor RPMs

Speed (RPM)	f_D (MHz)	v_{LOS} (m/s)		**PSD** (10^{-12} V^2/Hz)			**Peak PSD/ mean** (dB)	**Excess power** (10^{-5} V^2)
		Measured	Expected	Mean	Mean $+5\sigma$	**At f_D**		
50	0.074	0.057	0.056	1.15†	1.57	294.1	24.1	3.65
100	0.162	0.126	0.111	1.38†	1.98	421.3	24.9	3.65
250	0.397	0.308	0.278	1.20†	2.09	465.3	25.9	3.57
500	0.793	0.615	0.555	1.68†	3.99	257.9	21.9	3.58
1000	1.577	1.222	1.111	1.84	3.03	126.2	18.4	3.61
2000	3.174	2.460	2.221	1.39	2.54	68.0	16.9	3.50

Table 5.1: C: PSD Doppler peak and noise statistics for each motor RPM tested

† Peak too close to DC for a full noise band below it; the floor is taken mostly or entirely from the high-frequency side and is likely a slight underestimate.

A distinct Doppler peak was obtained at every speed, between 16.9 dB and 25.9 dB above the local noise floor and well clear of the mean $+5\sigma$ detection threshold in every case, satisfying P4. The peak height falls and its width grows with speed because the velocity spread across the beam spot scales with N. However, the integrated excess power stays constant at $(3.6 \pm 0.1) \times 10^{-5}$ V^2, which shows that the returned power is unchanged and is merely spread over a wider band. At 50 RPM ($v_{\mathrm{LOS}} = 0.056$ m/s) the peak is still clearly resolved, satisfying P5; here however, the 10 kHz bin width of the Welch PSD (corresponding to 7.75 mm/s of velocity) and the proximity to DC limit the precision, which explains why this point does not show the systematic error seen at higher speeds, as discussed below.

It can be seen that f_D, the centroid of each Doppler peak, and its corresponding measured velocity, $v_{LOS} = \frac{f_D}{1.29MHz}$, was higher than the predicted Doppler peak, by approximately

10% for each case, causing the system to fail performance requirement P7 for the hard-target case. This is illustrated visually in Figure 5.3:

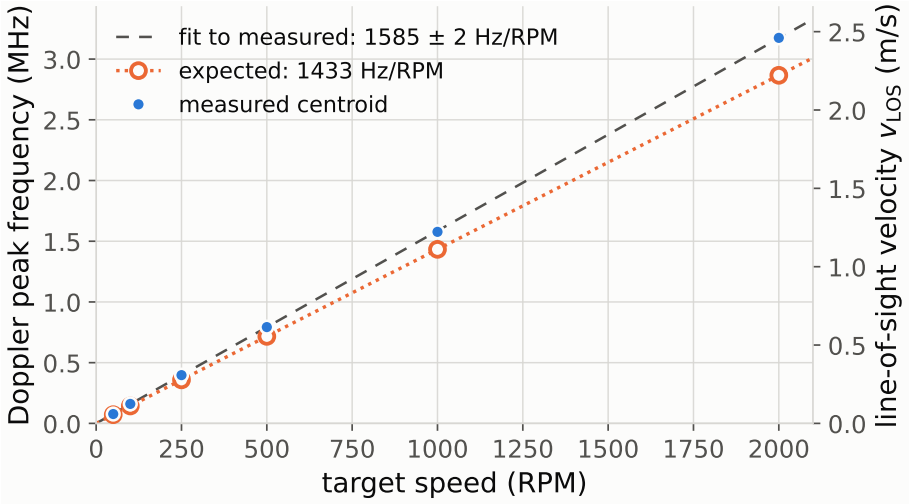


Figure 5.3: C: Measured Doppler peak frequency (from centroid) and expected Doppler peak frequency vs target motor RPM

As Figure 5.3 shows, f_D is linear in N: a fit through the origin gives 1585(2) Hz/RPM, and all points from 250 RPM upwards lie within 0.5 % of it. However, the slope is 10.6 % above the expected 1433 Hz/RPM, so P7 is not met as built. Because the error is a constant scale factor, it points to the geometry rather than to the measurement itself. Since v_{LOS} is proportional to d, the beam striking the disc only 1.6 mm further from the axis than intended would account for it. Over the 100 mm collimator-to-disc distance, that is a collimator tilt of only about 0.9°. Adjusting the collimator bracket screw changed the percentage error between experiment sets, while it remained constant within each set, which supports this explanation.

In summary, the experiment validates the Doppler measurement principle: the system resolves speeds from 0.06 m/s to 2.5 m/s, with a linear response and a large SNR margin on a hard target. The 10.6 % offset is a calibration error in the rig geometry, which a single-point calibration would remove. The aerosol return in experiment D is expected to be far weaker, so this margin does not by itself guarantee wind-speed measurement, which is addressed in section 5.2.

5.2 D: Wind-speed Measurement

5.2.1 Experimental Goals

Now that the noise analysis of the system is complete, and the speed-measurement principle has been shown to be functional for a hard-target, the system can finally be tested for seeded air wind-speed measurement. The primary subject of investigation for this section is the sensitivity of the system with respect to its noise and the different levels of seeding in the air, as well as its accuracy at different tested speeds. As discussed previously, the range of the system can not be tested in a wind-tunnel environment, and thus this is out of scope for this project.

5.2.2 Experimental Methodology

In order to test wind-speed measurements in a controlled environment, a test-setup was built in an existing wind-tunnel. The wind-tunnel is capable of producing measured, stable flows at 3 - 21m/s, at the centre of the test section. The tunnel has an inlet, where an air seeding device can be placed to seed the tunnel air. Two types of seeding are investigated; "fog" machines, which use mixtures of glycol or glycerin, and distilled water to produce a dense, visible vapour cloud in the air (of the same type typically seen in entertainment venues), and "misters", which simply use water to generate a very fine (barely visible) mist in the air.

The test setup used in the wind-tunnel is shown in Figure 5.4:

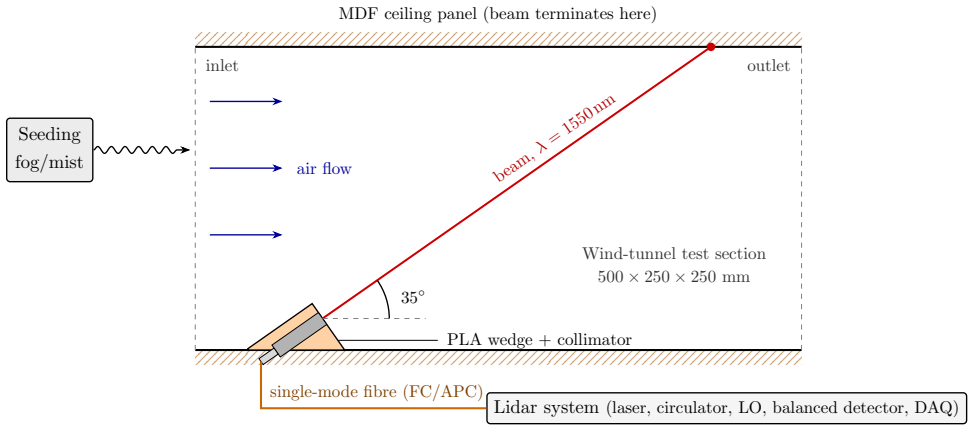


Figure 5.4: D: Wind-tunnel experimental setup diagram

A small wedge which fits the collimator and which can be attached to the floor of the

wind-tunnel was designed and 3D printed. The collimator is pointed downstream of the flow, so as to minimise any residue buildup from seeded air hitting the collimator lens from front-on. The consequence of this however, is that the collimator has to be placed at an angle (35 degrees in this design), rather than straight in the direction of flow, so as to avoid the collimator and the housing from disturbing the flow and causing excessive turbulence, immediately prior to the measurement line-of-sight. The angled wedge also provides the advantage of allowing the beam to terminate on a stationary angled surface (the MDF ceiling of the wind-tunnel), rather than continuing on through the tunnel to hit the rotor which generates the flow, which would generate additional Doppler components at the radial velocity of the rotor-blades in the direction of the beam. The angled beam termination surface also serves to absorb and reflect unwanted stationary (non-Doppler shifted) beam components away from the system, so that there is minimal additional PIIN - this mimics the setup of section 4.3.

It must be noted from the rig geometry, that the system only measures the component of the velocity which is in the direction of the beam. Assuming no vertical flow in the tunnel, the velocity "seen" by the beam is:

$$v_{los} = cos(35) \times U \tag{5.2}$$

Where U is the horizontal flow-speed in the wind-tunnel, as measured by its sensor. As previously calculated, a 1550nm beam will be Doppler shifted by approximately $1.29MHz$ per m/s of flow in the direction of the beam. Thus, for this test setup, a Doppler shift of

$$\Delta F = 1.29MHz \times cos(35) \times U = 1.057MHz \times U \tag{5.3}$$

is expected, i.e. $1.057MHz$ per m/s of wind-tunnel flow.

Again, the system is fully set up for these experiments as for C, as shown in Figure 3.2 and with a laser drive current of 48mA. The Welch method for calculating PSD was again used here, as for experiments A, B, and C, however in this case 500 oscilloscope records are obtained, and each segment length is set to 2500, giving a total of approximately 7200 independent averages (see section 3.5). This gives a lower noise variance which will make the detection of a smaller Doppler peak easier, in the absence of a laser power amplifier which would increase the aerosol return, and thus increase the size of the Doppler peak.

Even with this high amount of averages, the 500 records of 25 000 samples at 125 MSa/s represent only 100 ms of sampled data, which is comparable to the 4 ms to 41 ms integration times of established CW lidars[13][4]. Each 200 µs record is short enough that the flow is effectively frozen within it, so with a continuously streaming digitiser, with an FPGA (or similar) to process the data in real-time, the same averaging would give

a new velocity estimate every 0.1 s. This 10 Hz update rate is well above the 0.5 Hz to 1 Hz found to be sufficient for lidar-assisted wind turbine control [22], though below the 50 Hz of the commercial ZephIR [13]. In this set-up, however, the oscilloscope's transfer bottleneck spreads the records over roughly 60 s, so each PSD is an average of the flow over that period. This is acceptable here because the tunnel speed is held constant during each capture, but in a field-deployed iteration of the system an FPGA based acquisition chain should certainly be investigated.

5.2.3 Results and Discussion

Sensitivity and SNR

Figure 5.5 below shows the PSD of the data obtained for different wind-tunnel setpoints, with dense fog-seeded air. It can be seen that there is a large variation in the noise floor between the experiments, and even within single experiments (presenting as a slope in the floor's curve), making it difficult to clearly distinguish the different curves, thus a second plot is provided alongside the raw PSD, where each PSD is normalised by its respective noise floor, so as to make the results more clear. Normalisation is done by dividing the PSD by its own running median (61 bins). Table 5.2 shows the location of the Doppler peak, f_D, for each wind-tunnel setpoint, as well as the noise PSD statistics, the PSD at f_D, and the excess power from each peak (calculated by integrating the peak across the frequency it spans). Note that the data in Table 5.2 is also taken with the PSD normalised to the noise floor. See Appendix H for the raw PSDs of each case plotted on their own axes, and full tables of raw results.

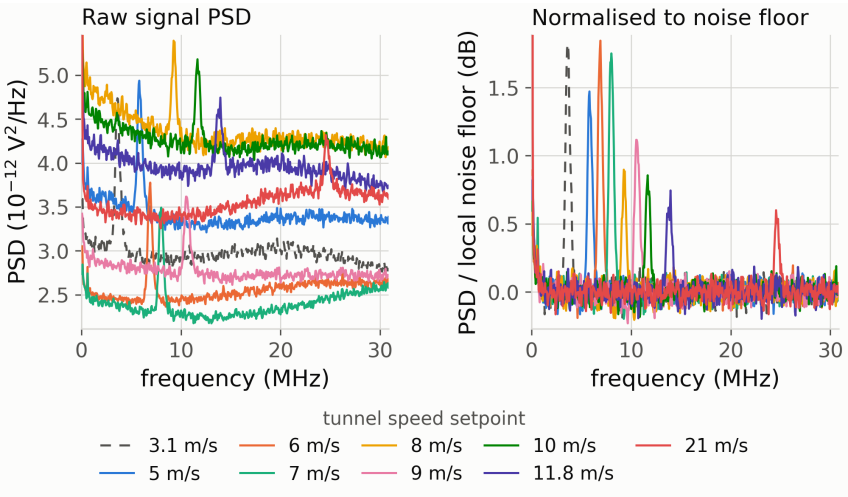


Figure 5.5: D: PSD for wind-speed measurements, at different wind-tunnel setpoints, with dense fog-seeded air

Speed (m/s)	f_D (MHz)	**PSD/floor** (dB)		**Excess** ($10^{-7}\,V^2$)
		Mean $+5\sigma$	**At f_D**	
3.1	3.637	0.226	1.815	10.25
5.0	5.868	0.295	1.340	8.87
6.0	6.892	0.217	1.843	7.55
7.0	7.997	0.304	1.749	7.04
8.0	9.280	0.320	0.868	5.99
9.0	10.611	0.322	1.099	6.08
10.0	11.664	0.214	0.855	6.00
11.8	13.837	0.254	0.526	5.27
21.0	24.674	0.262	0.502	4.01

Table 5.2: Normalised PSD of Doppler peak relative to noise for dense fog-seeded air. Full raw statistics are given in Appendix H.

It can be seen that at every speed tested in dense fog, a clear Doppler peak was detected above the mean $+5\sigma$ threshold, with a test setup which gives less than 10 000 averages, thus fulfilling performance requirement P6. It can also be seen that at higher speeds, the excess power at the Doppler peak decreases; this is because the seeding becomes more diffuse in the air at higher speeds, thus making the returned signal smaller, and making the peak clear the detection threshold with less margin.

After measuring the system's sensitivity performance in the dense-fog seeded air, the system was tested using the same test setup, but this time using very light mist seeding, which is more akin to real-world atmospheric conditions. Since the significantly lighter seeding gives a smaller return signal, particularly in faster wind-tunnel speeds, more than one capture is obtained at each speed, so that more averages can be taken. A capture here is defined as 500 oscilloscope records, as discussed in the methodology. The results of this variation in the experiment are detailed in Table 5.3, and Figure 5.6 shows the PSD near the peak for different numbers of averaged captures (e.g. 2 captures averaged = 1000 records averaged = 14 400 total independent averages after taking Welch PSD as described in section 3.5).

Speed (m/s)	**Captures averaged**	**Normalised PSD** (dB)		**Excess** ($10^{-7}\,V^2$)
		Mean $+5\sigma$	**At f_D**	
5.0	1	0.255	0.141	4.38
5.0	2	0.131	0.114	4.45
5.0	3	0.116	0.160	3.75

Table 5.3: Normalised PSD of Doppler peak relative to noise for light mist-seeded air. Full raw statistics are given in Appendix H

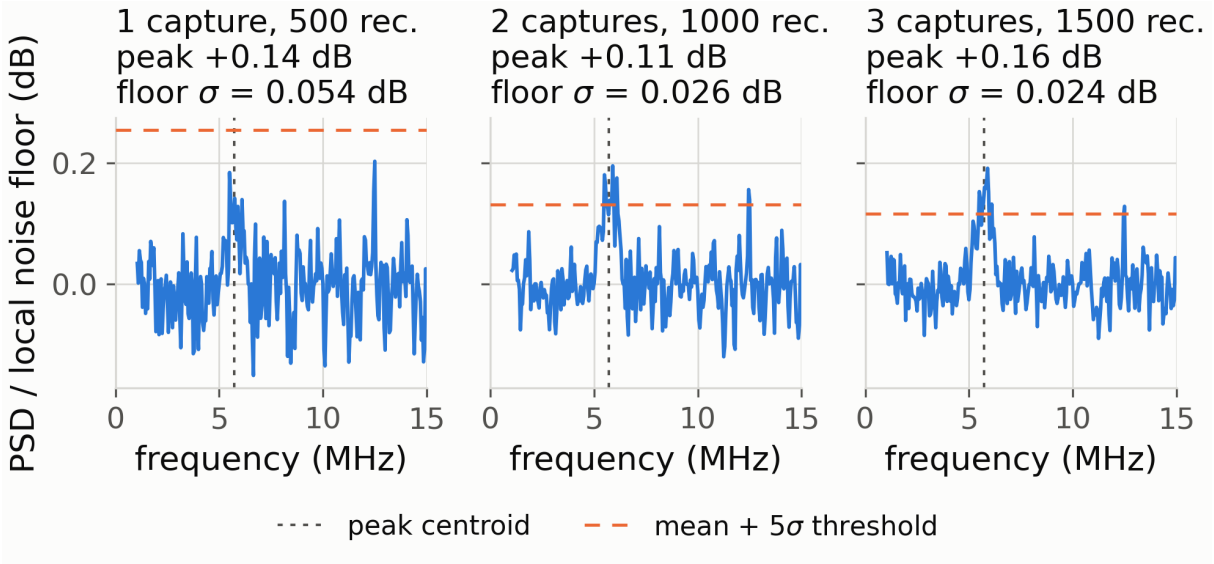


Figure 5.6: Normalised PSD near f_D for different numbers of averages

It can be seen in Table 5.3 that at 5 m/s the light mist gives a substantially weaker signal return than dense fog at the same tunnel speed: its excess power is approximately half the $8.87 \times 10^{-7}\,\mathrm{V}^2$ measured in fog (Table 5.2), a reduction of about 3–4 dB. As a result, with a single 500-record capture (about 7200 independent averages, see section 3.5) the peak lies below the mean $+5\sigma$ detection threshold. With two captures it becomes visible in Figure 5.6, but its value at the centroid (0.114 dB) still falls just short of the 0.131 dB threshold, making this a marginal case. With three captures (about 21 600 independent averages) the peak clears the threshold unambiguously. Figure 5.6 shows that averaging does not raise the Doppler peak, which remains at approximately 0.14 dB (its variation between captures is within the noise of a single bin), but reduces the fluctuation of the noise floor, lowering the threshold from 0.255 dB to 0.116 dB and so increasing the size of the peak **relative** to the noise. This illustrates how averaging increases the sensitivity of a system that, lacking an optical power amplifier (section 3.4), has a weak aerosol return.

Although the light-mist case does not meet P6 within 10 000 averages, it illustrates both the benefit and the limit of averaging. At 5 m/s the Doppler peak lies only ≈0.14 dB (3.3 %) above the noise floor, whereas the mean $+5\sigma$ threshold at $N = 10\,000$ is 5 % above it. A further ≈2 dB of signal-to-noise ratio is therefore needed to meet P6, consistent with the ≈23 000 averages found necessary in practice. However, because $\sigma \propto 1/\sqrt{N}$ (see section 3.5), each doubling of N gains only 1.5 dB, so averaging cannot practically close larger gaps, and measurement in unseeded, clean air is such a gap:

The current laser power emitted is estimated at approximately 15mW - from the 17mW at the diode output for a current of 48mA (see result of experiment A2), only 90% is directed at the atmospheric emission, and power loss in the fiber is ignored here. Converting the

integrated excess power of the 5 m/s peak to optical power using Equation C.4 gives an effective received power here of roughly 0.1 pW, from the 15mW emitted power. In open air, where the beam is not cut short by the tunnel ceiling and Equation 3.1 applies, clear air ($\beta(\pi) \approx 10^{-8}\,\mathrm{m^{-1}sr^{-1}}$ [13]) would return only ≈0.7 fW from the transmitted beam by this system in its current form, which is 22–25 dB lower than the light mist-seeded case. A clean-air measurement would therefore require roughly 25–30 dB of signal increase or noise reduction, equivalent to about 10^5 times more averaging. This is approximately what an erbium-doped fibre amplifier (see discussion on laser power level in section 3.4) raising the transmit power to 1 W (+18 dB above 15mW), combined with removing the residual PIIN to reach the shot-noise limit (+5.7 dB, section 4.3), would provide. Thus, these measurements in seeded air give a clear upgrade-path to the system, and are able to quantify how much more sensitivity is required in the system in order to measure wind-speed in unseeded air. It also illustrates that simply taking more averages to increase the sensitivity is not viable to achieve wind-speed detection in unseeded air.

Velocity accuracy

Table 5.4 gives a summarised set of results for the velocity accuracy of the DWL measurements, after converting the detected peak centroids from a frequency to a velocity using Equation 2.2. A full table of the peak centroid-frequencies and corresponding velocities compared to the wind-tunnel setpoint can be viewed in Appendix H.3.

Quantity	**Value**
Tunnel speed range	3.1–21 m/s
Speed error	+10.7 %
Error per speed	+8.1 to +11.6 %
Linearity, R^2 (free intercept)	0.9998
Speed error after one-factor calibration, RMS (max)	0.08 m/s (0.17 m/s), 1.1 %
Doppler peak FWHM	0.45–0.65 MHz (0.35–0.50 m/s LOS)

Table 5.4: Velocity accuracy in dense fog (collimator facing downstream, 9 speeds).

It was found that using the original test setup detailed in the methodology for this section (downstream facing collimator and wedge), an average overshoot of approximately 10.7% was consistently observed between the DWL measured velocity (as calculated from the peak's frequency), and the wind-tunnel's velocity setpoint. However, the data showed very good linearity, with $R^2 = 0.9998$, thus the overshoot offset can easily be removed via a single-point calibration, to give an error of approximately 1.1%. Without this calibration, the system fails performance requirement P7 on velocity accuracy within 5%, but after

the calibration it passes. Thus, it is important to investigate whether the calibration, which assumes error from the test setup, is valid.

The cause of the 10.7% overshoot was therefore further investigated: The laser's frequency is stable at the expected 1550nm (a drift causing 10% error would require a drifted wavelength of 1400nm, which is not possible for a DFB diode), the data acquisition chain was validated against a signal generated by a signal generator, and the wind-tunnel's setpoint speeds were validated against a calibrated anemometer. Thus it is hypothesised that this error is due to the turbulence and blockage caused by the test-rig geometry, which may redirect a significant component of the flow over the top of the wedge, and thus into the line of sight of the collimator, so as to cause an overshoot in its measurement relative to the wind-tunnel setpoint. The test setup was modified to investigate this, so as to have the wedge and collimator now pointing upstream of the wind. The results of this setup are plotted in Figure 5.7 alongside the "expected" measurement from the wind-tunnel setpoint, and the results from the original downstream setup. Note here that the system is homodyne, and thus downstream and upstream velocity appear the same at the output (the recovered velocity is not signed, see section 2.4).

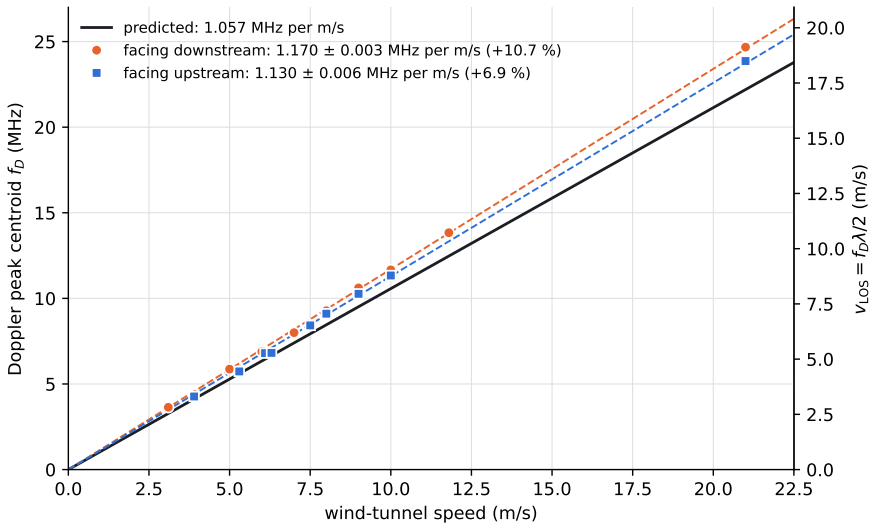


Figure 5.7: Measured vs predicted velocity at different wind-tunnel setpoints

It can be seen that with this new setup, the overshoot was reduced to an average of 6.9%, while retaining the linearity of the original setup. It is thus clear that at least some component of the overshoot is due to the test-rig geometry, and that it is thus valid to remove the offset using a single-point calibration. Rigorously profiling the tunnel flow for a given test rig would however be needed to test this hypothesis, and the velocity accuracy further (see recommendation R4).

Chapter 6

Conclusions and Recommendations

This report has demonstrated a working benchtop-scale Doppler wind LiDAR, built from readily available components. The conclusions from each experiment are given first, followed by recommendations for improving the system; each conclusion refers to the recommendations that address it.

6.1 Conclusions

Firstly, it can be seen in the noise analysis that while the system was able to achieve shot-noise limited operation with respect to the electronic noise of the detector (thus meeting requirement P1), this was not maintained once PIIN was introduced. The final system was PIIN limited; after matching fiber path lengths, internal PIIN was measured at an average of approximately 5.7dB above the shot-noise level, with significant variation and instability in this noise being seen across measurements. Prior to matching fiber path lengths, PIIN was measured at 22.1dB above shot noise. This illustrates the importance of path-length matching in reducing PIIN to acceptable levels, but also highlights that despite matching the path-lengths, the system is still 5.7dB above the optimal sensitivity which can be reached, which is the shot-noise limit. See recommendation R1 for suggested improvements on this.

Thereafter, the functional analysis yielded several important results. It was firstly verified that the system is capable of hard target velocimetry, with Doppler peaks detected at 16.9 - 25.9dB above the noise floor, at less than 1000 PSD averages (thus meeting P4). While the wind-tunnel could not be used for testing the minimum resolvable LOS speed, a hard-target speed of 0.056m/s was measured in Experiment C, thus meeting the low-end requirement of P5.

After validating the system's working principle on a hard-target, testing of wind-speed measurement was conducted. It was found that the system produced measurable Doppler peaks above the detection threshold at all tested wind-speeds between 3 - 21m/s, for dense fog seeded air, at less than 10 000 PSD averages, thus meeting P6 and the high-end

requirement of P5. The time taken to acquire this amount of averages to obtain a result is shown to be in line with acquisition times in commercial systems when the oscilloscope's bottleneck is ignored, however recommendation R2 discusses the implementation of a data acquisition chain which does not have this same bottleneck. The system was also tested using lighter mist-seeded air. For this test case, it was found that the returned signal power was far lower, and that the system only reached the detection threshold after approximately 21 600 independent PSD averages. This illustrates the benefits of increased averages in lowering the detection threshold by lowering noise variance. However, crucially, it was found that the system built as is would require an impractical 10^5 times more averages for peak detection in unseeded air with yet still lower signal return power. This is attributed to the lack of an optical power amplifier in the system, and the high PIIN. Recommendation R1 and R3 discuss improvement paths relating to this.

It was also found that the system's Doppler peak measurements, when converted to velocity, overshot the expected velocity values relative to the wind-tunnel setpoint by approximately 10%, thus failing requirement P7 on velocity error being within 5%. Further experimentation however showed that this error could partially be attributed to the test-rig geometry, and could be reduced via a single-point calibration. After the calibration, the error was approximately 1.1%, and thus within the bounds to meet P7. Recommendation R4 discusses this matter further.

Ultimately, the contents of this report effectively demonstrate the working principles of a Doppler wind LiDAR system and quantifies its detection sensitivity and velocity accuracy with respect to the noise in the system and the components used.

6.2 Recommendations

From the concluded results discussed above, a number of recommendations can be made for future work on this system. These recommendations form an essential part of the project's work, as they are derived from quantified limitations in the current system.

R1 Narrow linewidth laser source: It is recommended that a narrower linewidth laser source in the order of 50 - 100kHz is obtained. This serves to reduce PIIN, which currently raises the noise floor to approximately 5.7dB above the shot-noise floor, and which is highly unstable, so that the total noise of the system is reduced to approximately 0.6dB above the optimal shot-noise level. This will increase the detection sensitivity of the system, and increase its ability to resolve turbulence via a narrower Doppler peak.

R2 High sample-rate ADC and FPGA processing chain: The current setup uses an oscilloscope and laptop as the processing chain. While the oscilloscope can sample at a high enough rate, it has a data transfer bottleneck which results in measurements being taken across a time period of 30 - 60 seconds, with only approximately 0.2% of data in this period actually being sampled and used. It is thus recommended that instead an ADC and FPGA processing chain can be developed, which samples continuously and performs the necessary FFT calculations at a high rate. This can increase the data usage for a given time period up to 100% [13], thus reducing the acquisition time to be in the order of 100ms, and giving the system a sample rate of up to 10Hz, which is in line with commercial systems and real-world wind conditions.

R3 Optical power amplifier: The system as is does not emit a very high power laser beam (approximately 15mW), and the returned backscatter power is thus very low, limiting the system's ability to measure Doppler peaks in clear, unseeded air. It is recommended that an optical power amplifier (ideally an EDFA) be added so as to increase the emitted power, and thus increase the system's sensitivity and detection ability.

R4 Test environment flow-profiling: It is recommended that a wind-tunnel test environment's flow is comprehensively profiled, and that a test-rig which minimally interferes with this flow is designed, so that the accuracy of the system can be better tested and quantified and does not rely on calibration for accurate measurements in a controlled environment.

R5 Focusing optics: It is recommended that focusing optics are added after the collimator lens. This would allow the system to achieve ranging of the measured wind-speeds, rather than averaging the flow equally over the entire beam.

R6 Acousto-optic modulator: It is recommended that an acousto-optic modulator be added to the system. This converts the system to a heterodyne, rather than homodyne detection system, and would thus allow for the resolution of the direction of the velocity, rather than just the magnitude. This is essential for real-world applications where the flow direction is not known, as it was in this report's test environment.

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Appendix A

Graduate Attributes

The table below summarises how the graduate attributes (GAs) associated with this course have been met, and where in the report the evidence can be found.

Table A.1: Graduate attributes and where they are addressed in this report

GA	Requirement	Justification and section in the report
1	Problem-solving	My project concerns the field of LiDAR for wind-speed measurement. Specifically, my project deals with the problem of designing a prototype benchtop-scale version of a Doppler wind LiDAR system. I consulted various sources of literature in order to first understand what the key use-cases and goals of Doppler wind LiDAR systems are (mostly with respect to wind-turbines, and the use of such systems to control the turbines optimally). Thereafter, with the end-use in mind, I reviewed sources which discuss common types/classes of such systems (chapter 2), and identified a system type which would be suitable for this project, given the time-frame and budget – this type being coherent detection Doppler LiDAR (section 3.2). After identifying this suitable class of system, I reviewed the various design considerations of such a system in the available literature (chapter 2). This included mathematical and physical descriptions of how the system operates, with respect to signal theory, optics, electronics, and relevant physics phenomena (including Rayleigh and Mie scattering, coherent detection methodology, balanced detection, and laser source physics). From this, I synthesised a list of parts which would be suitable for use in the project, and used the relevant mathematics to validate these choices in terms of the final requirements (section 3.4, section 3.3).

Continued on next page

Table A.1 (continued)

GA	Requirement	Justification and section in the report
4	Investigations, experiments and data analysis	I conducted a thorough investigation into the various types of Doppler wind LiDAR systems which are used in industry and scientific research. After doing so, I conducted more specific research into coherent detection Doppler LiDAR systems, and the physics, signal processing and optics which underpin these systems (chapter 2, section 3.4). From this, I was able to recommend to my supervisor a course of action for the class of system to be designed, and substantiate this choice through the research done. I formulated various experiments to conduct on the system once it was built, which validate the signal theory, mathematics and electronics principles concerned, and validate the system's workings (chapter 4, chapter 5). The primary consideration in these experiments is a thorough noise analysis (chapter 4), which serves to quantify the sensitivity of the system. Furthermore, the effects of various design decisions on the performance (e.g. source laser linewidth and power, or the use of balanced detection over single-ended detection) are analysed in these experiments, and optimal requirements for such systems are thus experimentally shown and discussed, and suitable recommendations for upgrades are made.
5	Use of engineering tools	Engineering mathematics was used in the system's design in order to determine the optimal components to use (e.g. the fibre-optic splitter ratio in section 3.4). Reasoning based on the reviewed literature was applied to predict performance outcomes and to determine how the system should be tested (e.g. the prediction that the system will require seeded air for wind-speed measurement due to the lack of optical power amplification, section 3.4). Once the system was built, an oscilloscope was used in conjunction with Python scripts to acquire data from the system (setup described in section 3.5, used in chapter 4 and chapter 5). The data was also post-processed using Python scripts so as to gain useful information from it, and to generate plots and tables which show the obtained data in a readable manner.

Continued on next page

Table A.1 (continued)

GA	Requirement	Justification and section in the report
6	Professional and technical communication (long report)	I have attended scheduled meetings (and scheduled further follow-up meetings) with my supervisor in order to discuss the route which the project should take, as well as budget and technological considerations. I have also consulted with Maysam Soltanian to obtain suitable sources for components which are required for the project. This written project report serves to communicate the technical details of the system's development and testing to an engineering audience. The abstract and conclusion of this report serve to communicate basic ideas (e.g. the need for system upgrades in order to perform wind-speed measurements in clean air) to a non-engineering or management audience.
8	Individual work	I have maintained goals and suggested timeframes for my work on the project. While meetings with my supervisor and Maysam helped guide the sourcing and budget constraints for the project components, all of the specific components were selected by me, and suitable sourcing suggestions were made to Maysam. While the supervisor provided the overarching topic of Doppler wind LiDAR, I used my own research into the area to select an appropriate subclass of LiDAR system to build for the project, synthesising the constraints of the project and the desired requirements. All experimental work was synthesised individually. Appendix I shows a knowledge of professional ethics and norms in its declaration of how and where artificial intelligence was used in this project.

Continued on next page

Table A.1 (continued)

GA	Requirement	Justification and section in the report
9	Independent learning ability	I have independently learnt concepts in photonics (laser source characteristics), atmospheric sciences (Rayleigh and Mie scattering) and signal theory (coherent detection and balanced detection), in order to understand and synthesise the design of a Doppler wind LiDAR system. The initial problem was ill-defined and the decision to take the route of designing a benchtop-scale Doppler wind LiDAR system was not immediately obvious. It was only after learning more about the field and evaluating areas of research in it that this decision was made, after reaching out to my supervisor to consult on this. Throughout the project I have had to manage the scope of the project to suit my abilities and constraints, given that the field of research contains many complexities which are beyond an undergraduate skill-level and time-frame. This therefore shows my ability to synthesise meaningful objectives, knowledge and outcomes from an ill-posed problem.

Appendix B

Linewidth, PIIN and Doppler Peak Broadening

In order to understand the linewidth property of a laser source, it is important to first have an understanding of the concept of phase noise in a laser:

An ideal laser emits an electromagnetic wave $E(t) = E_0 cos(2\pi f_0 t + \phi)$ with ϕ a constant. A real laser emits $E(t) = E_0 cos(2\pi f_0 t + \phi(t))$ where $\phi(t)$ drifts unpredictably. This drift is called phase noise. Thus, when considering the frequency spectrum of the emitted laser beam over a long time period, it takes a Lorentzian shape centred around the nominal frequency of the laser (whereas an ideal laser would present simply as a delta spike at the nominal frequency). The laser linewidth, Δf, is then defined as the -3dB bandwidth of this Lorentzian spectrum, as illustrated in Figure B.1 below:

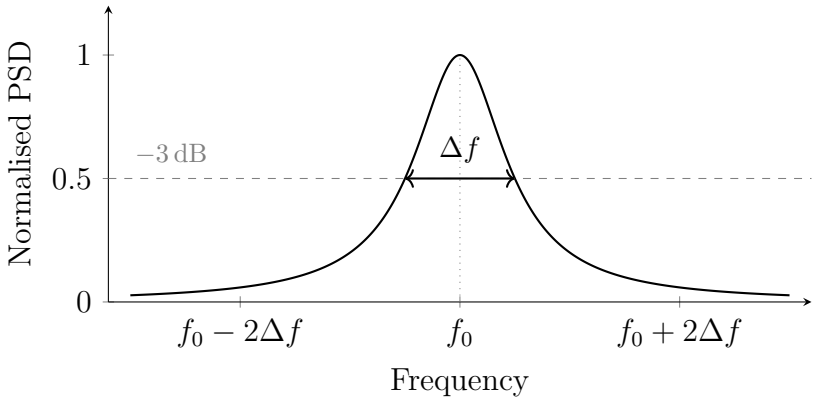


Figure B.1: Lorentzian frequency spectrum of a laser diode.

Asssociated with the linewidth is a quantity termed coherence length. The coherence length of the emitted beam, L_c, is given by [4]:

$$L_c = \frac{c}{2\pi\Delta f} \tag{B.1}$$

Where Δf is the linewidth of the source laser, and c is the speed of light. The coherence length is of relevance when considering linewidth effects.

The non-ideal linewidth property of a laser source gives rise to two primary effects in a DWL system's measurements - namely phase-induced intensity noise (PIIN), and the broadened Doppler peak:

As noted by Cariou et al. [6], Jiang et al. [14], and Harris et al. [12], reflections of the outgoing beam - caused by optics and the fibre optic transmit path - travel back in the system and are interfered with the reference beam (LO) before detection (in the same way as the actual backscattered signal of interest is). Because the reflection and LO derive from the same source, the mixing product is a strong, unwanted tone at baseband (homodyne) or at the intermediate frequency (heterodyne), carrying the phase noise accumulated over the path-length mismatch ΔL between the reflection and the LO. Harris et al. [12] show that the form of this residual noise is set by how large ΔL is relative to the coherence length (L_c) of the laser source.

For $\Delta L \gg L_c$ the two phases are highly uncorrelated because the difference in path (and therefore time) taken has allowed the phase of the two to "wander" significantly - and thus the tone broadens into a Lorentzian of width $2\Delta f$. This case is illustrated in Figure B.2 below, where it can be seen that this unwanted Lorentzian peak can serve to swamp out the desired Doppler peak to be measured - this is especially impactful for low wind-speeds, where the Doppler peak will be closer to the unwanted common-mode Lorentzian peak, and will be hidden more by it.

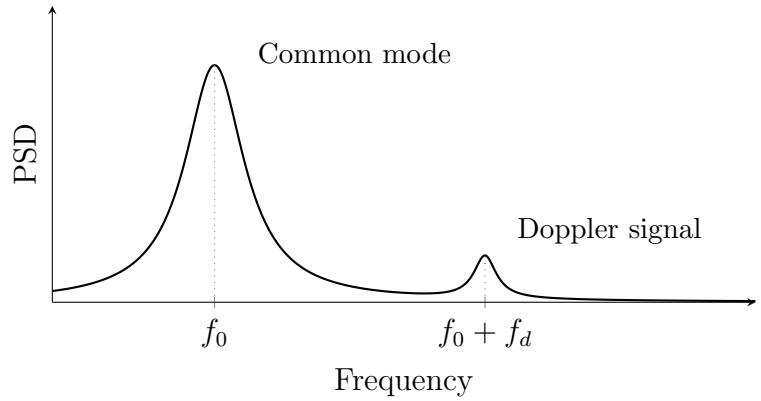


Figure B.2: Spectrum of a monostatic CW coherent lidar, showing the Doppler signal superimposed on the skirt of the common-mode peak (adapted from [6]).

For $\Delta L \ll L_c$, the case where the mismatch between LO and reflection paths is smaller, the reflected signal and the LO will be more correlated in phase. Thus, the residual instead appears as a flat excess noise floor across the frequency band, of level proportional to $\Delta f\, \Delta L^2$ [12]. This excess noise is termed excess LO noise, interferometric noise or phase-induced intensity noise (PIIN), depending on author [12, 1, 14]. For the remainder of this paper, this excess noise will be termed PIIN. PIIN has been measured at 5 - 30dB above

shot noise in a diode-laser lidar [12] and well above shot noise in a fibre-laser system [14]. Thus, PIIN is a significant noise source which can cause the system to operate far from the ideal shot-noise limit discussed in section 2.5. It is thus clear that to reduce PIIN, a laser with as small a linewidth as possible should be selected, and the path length mismatch, ΔL between LO and reflection paths should be minimised.

Another consideration arising due to non-ideal laser linewidth is the broadening of the measurable Doppler peak. This is particularly notable for longer ranges.

If the coherence length from Equation B.1 is long relative to the emitted signal's path length, the broadening of the Doppler peak due to the laser's linewidth may be neglected, and it will instead be other factors (such as turbulence in the air, and speckle) which dominate and cause a broader measurable peak [4]. This is because the total round-trip time of the beam is shorter, thus ensuring that the phase of the beam has not had "time" to shift significantly in between transmission, and later reception and mixing with the local oscillator. Thus, the received backscatter signal and the LO phase will be more coherent and will result in a sharper peak after mixing.

Conversely, if the coherence length is short relative to the signal path length (i.e. for targets at greater ranges), the Doppler peak will be spread over $2 \times \Delta f$, meaning that the measurement resolution and the ability to resolve turbulence in the air will be reduced [4].

From the two effects described above, it can clearly be seen that a narrow linewidth laser source is highly beneficial for DWL systems, in order to achieve the best-case shot-noise limited operation, and to achieve a narrow, measurable Doppler peak.

Appendix C

Balanced Detection

As discussed in section 2.5, the best-case performance of a DWL system is for it to operate in a shot-noise limited regime - thus, all other sources of noise should be reduced to below the shot-noise level. In coherent detection LiDAR architectures, the use of a balanced detector setup has become common for this purpose, as it assists in reducing a laser source's relative-intensity noise (RIN) below the shot-noise level, in a manner which single-ended detection does not.

A single-ended detection system simply uses a 2:1 (2 inputs, one output) fiber optic coupler to mix the LO reference signal with the backscatter signal, and uses a single photodetector to convert this light signal into a measurable current. It can be seen that this detection method retains the DC terms P_{LO}, P_{RS}. Any RIN from the laser source is present in the output, as these intensity fluctuations are contained in these two power terms. Thus, the measurable Doppler peak will be smaller with respect to the noise floor.

Contrastingly, balanced detection setup makes use of a 2:2 fibre optic coupler which also optically mixes the LO reference signal with the received backscatter signal (as discussed in subsection 2.3.2). However, the 2:2 coupler has two outputs, thus giving two output signals which can be termed E_1 and E_2 [15]:

$$E_1 = \frac{1}{\sqrt{2}}(E_{RS} + E_{LO}), E_2 = \frac{1}{\sqrt{2}}(E_{RS} - E_{LO}) \tag{C.1}$$

Where E_{RS} and E_{LO} are the complex electromagnetic fields of the returned backscatter signal and the local oscillator respectively.

Thereafter, the two signals enter the balanced detector, which consists of two matched photodiodes and a transimpedance amplifier to convert their currents into a measurable voltage. The photodiodes respond with currents I_1 and I_2 respectively, proportional to the intensity of the electromagnetic field incident on it ($|E_1|^2$ and $|E_2|^2$ respectively), as was also discussed in subsection 2.3.2. Due to the internal setup of the transimpedance amplifier and the photodiodes in the balanced detector, the output current is then given by [15]:

$$I_{det} \propto I_1 - I_2 \propto |E_1|^2 - |E_2|^2 \propto 2\sqrt{P_{LO}P_{RS}}cos(2\pi\Delta ft) \tag{C.2}$$

Comparing this to Equation 2.5, it can be seen that the DC terms P_{LO} and P_{RS} are no longer present due to them being common-mode in both I_1 and I_2 and thus being canceled by the subtraction. This therefore lowers the noise floor, as the RIN contained in these terms is also eliminated.

Writing in the photodiode responsivity R and the transimpedance gain G of the balanced detector, the output voltage of the detector is

$$V(t) = G\,(I_1 - I_2) = 2GR\sqrt{P_{LO}P_{RS}}\,\cos(2\pi\Delta f\,t), \tag{C.3}$$

where P_{LO} and P_{RS} are the total LO and received powers entering the coupler.

It must also be noted that since the mean of $\cos^2$ is $\frac{1}{2}$, the mean-square voltage of the beat signal is

$$\langle V^2 \rangle = 2G^2R^2P_{LO}P_{RS}. \tag{C.4}$$

Thus, by Parseval's theorem, $\langle V^2 \rangle$ equals the PSD integrated across the Doppler peak, so a measured excess power can allow one to estimate the received signal power: $P_{RS} = \langle V^2 \rangle / (2G^2R^2P_{LO})$.

Figure C.1 depicts the balanced detection process graphically:

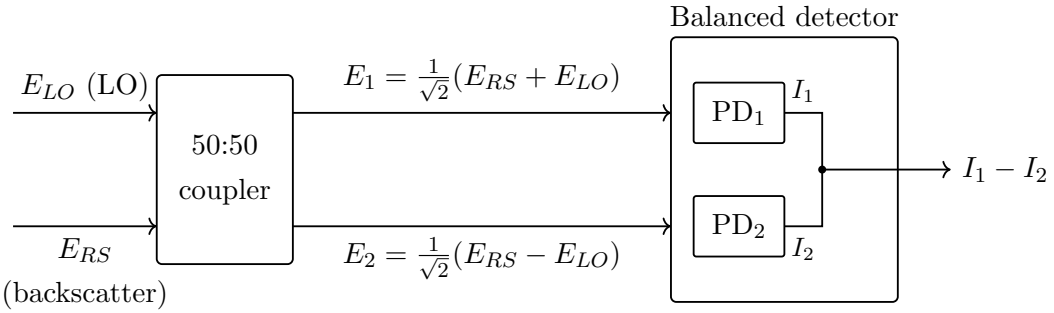


Figure C.1: Balanced detection. The 50:50 coupler delivers the sum and difference of the backscatter and LO fields to two matched photodiodes; the output is the difference of their photocurrents, in which the common-mode terms cancel and only the beat term $2\sqrt{P_{LO}P_{RS}}\cos(2\pi\Delta ft)$ remains.

It can thus be seen that balanced detection provides the advantage of cancelling RIN. This allows for the final detected signal to theoretically be able to achieve the desired shot-noise limited operation (if other sources of noise are also eliminated).

Appendix D

Laser Source Classes: Noise and Practical Notes

Source type	Noise and practical notes
Fabry–Pérot laser diode	Oscillates on several longitudinal modes simultaneously, so the output is not single-frequency and cannot be mixed coherently with a local oscillator; unsuitable for coherent detection [12].
DFB laser diode	Single-frequency, but with a linewidth of the order of MHz, which gives rise to significant PIIN in a CW system unless the LO and transmit path lengths are closely matched (section 4.3). Relaxation-oscillation peak at $\sim$1 GHz, far above the Doppler band, so RIN is flat and low in the frequency band of interest [6]. Supplied as a bare butterfly module; needs an external current driver and temperature controller. Selected for this project.
Planar external-cavity semiconductor	kHz-range linewidth, so PIIN and Doppler-peak broadening are greatly reduced compared to a DFB diode. Supplied with an integrated driver, and marketed specifically for wind lidar.
Er-doped DFB fibre laser	The common source of choice for CW anemometry owing to its very narrow linewidth [6, 13, 3]. Relaxation-oscillation peak near 1 MHz, however, so RIN is higher in the frequency band of interest than for a DFB diode [6].

Table D.1: Noise and practical notes for the laser source classes compared in Table 3.3.

Appendix E

Bill of Materials

Table E.1 gives the cost of each component in the final system design of Chapter 3. Prices are list prices at the time of purchase, converted to South African rand where necessary at approximately R18.4 per euro (October 2026), and exclude VAT, shipping and import duties.

Component	**Part / supplier**	**Qty**	**Cost (ZAR)**
Laser source			
DFB laser diode, 1550 nm, 20 mW	BFLD-1550-20SM-FA2	1	8 000
Laser diode driver	Koheron DRV200S-A-200 (EUR 315)	1	5 800
Butterfly mount	Koheron BTF100 (EUR 165)	1	3 040
Temperature controller	Koheron TEC100L-50 (EUR 345)	1	6 350
Fibre optics			
90:10 fibre splitter, SM, FC/APC	generic telecom	1	1 000
50:50 fibre coupler (optical mixer), SM, FC/APC	generic telecom	1	1 000
3-port fibre circulator, SM, FC/APC	generic telecom	1	1 500
FC/APC patch cords (incl. 2, 3 and 4 m LO-matching cables)	generic telecom	–	400
Optics and detection			
Fibre collimator, 1550 nm, $f = 37.13$ mm	Thorlabs F810APC-1550	1	5610
Balanced photodetector, 100 MHz	Koheron PD100B-AC (EUR 195)	1	3 590
Total			**≈ 36 290**

Table E.1: Approximate cost of the components in the final system. Prices exclude VAT, shipping and import duties.

Appendix F

Component Datasheets

The following sections give expanded specifications of the components used in this project. To see the full datasheets, visit the manufacturers' websites, or view the Github repository for this project at https://github.com/thomasbkruger/Final-Year-Project.

F.1 DFB Laser Diode (BFLD-1550-20SM-FA2)

The laser diode was supplied with the manufacturer's test report reproduced below (serial number 2630553009, tested 21 August 2026). General specifications: operating temperature 25 °C (15–35 °C); wavelength range 1550 ± 2 nm; output power ≥ 20 mW; operating current < 100 mA; single-mode fibre pigtail with FC/APC connector; 14-pin butterfly package.

Parameter	**Symbol**	**Measured value**
Centre wavelength	λ_p	1550.13 nm
Operating current (at 20 mW)	I_o	58.07 mA
Threshold current	I_{TH}	4.86 mA
Bias (forward) voltage	V_f	1.46 V
Monitor photodiode current	I_m	290 µA
Side-mode suppression ratio	SMSR	50.68 dB
TEC voltage	V_{TEC}	50 mV
TEC current	I_{TEC}	120 mA

Table F.1: Manufacturer test-report values for the BFLD-1550-20SM-FA2 laser diode (S/N 2630553009) at 25 °C.

Pin	Function	Pin	Function
1	TEC (+)	14	TEC (−)
2	Thermistor	13	Case ground
3	PD monitor anode (−)	12	NC
4	PD monitor cathode (+)	11	Laser cathode (−)
5	Thermistor	10	Laser anode (+)
6	NC	9	NC
7	NC	8	NC

Table F.2: Pin definition of the 14-pin butterfly package (thermistor 10 kΩ).

The test report also gives the measured P–I–V curve and the optical spectrum (1500–1600 nm), reproduced in Figure F.1.

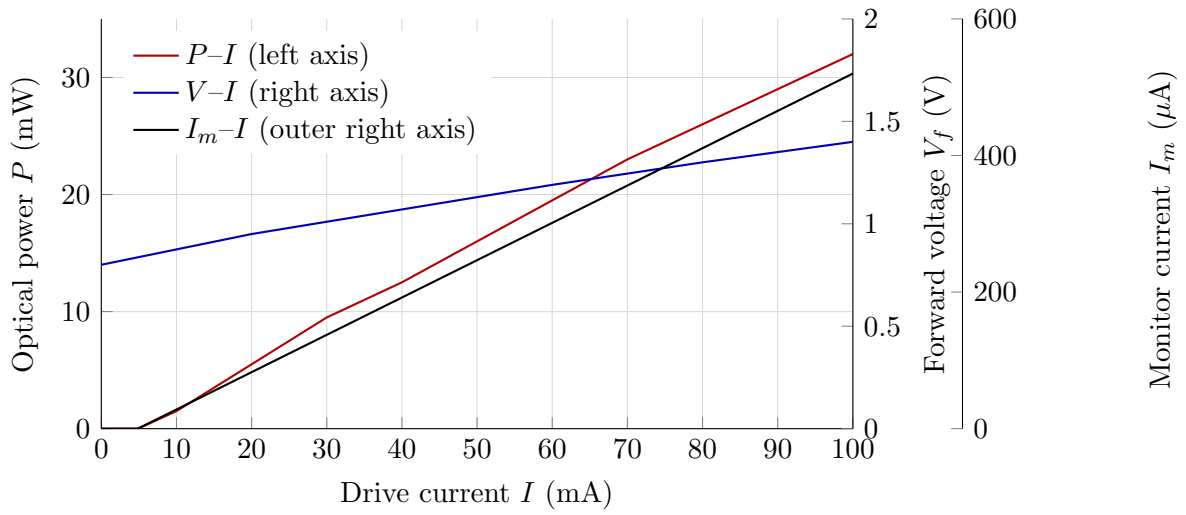


Figure F.1: P–I–V characteristic of the supplied BFLD-1550-20SM-FA2 diode at 25 °C, redrawn from the manufacturer's test report (values read from the printed curve to about ±0.5 mW).

F.2 Laser Driver (Koheron DRV200S-A-200) and Butterfly Mount (BTF100)

The DRV200S is an ultra-low-noise constant-current driver whose setpoint is controlled by an externally applied voltage, with an adjustable current limit and compliance voltage. The -A-200 variant was chosen to match the 58 mA operating current of the diode. It also provides a DC–6 MHz modulation input, which is unused in this project.

Parameter	**DRV200S-A-200**
Laser current range	0 mA to 200 mA
Current limit (adjustable)	65 mA to 300 mA
Compliance voltage	15.5 V (at $V_{SL} = 19$ V)
Current noise density (1 kHz)	270 pA/$\sqrt{\text{Hz}}$
RMS current noise (10 Hz–1 MHz)	265 nA
Temperature coefficient	30 ppm/°C
Current monitor gain	10 V A^{-1}
Modulation bandwidth (3 dB)	6 MHz
Supply voltages	internal 4.8 V to 6 V; laser 3 V to 19 V
Compatible lasers	floating diodes

Table F.3: Key specifications of the Koheron DRV200S-A-200 laser diode driver.

The diode is housed in a Koheron BTF100 mount for 14-pin butterfly packages (2 A per pin maximum, 51 × 42 × 32 mm), which breaks out the laser anode/cathode and their voltage-sense pins, the monitor photodiode, the thermistor and the TEC to a terminal block.

F.3 Temperature Controller (Koheron TEC100L-50)

The TEC100L is a compact high-stability temperature controller with a precision analogue front end and adjustable PID gains. The temperature setpoint is set as a thermistor resistance; the diode's 10 kΩ thermistor falls in the middle of its range.

Parameter	**TEC100L-50**
TEC current	−1.15 to 1.15 A
Setpoint (thermistor) range	5 kΩ to 20 kΩ (40 °C to 10 °C)
Compliance voltage at 1 A	±4.5 V
24-hour stability	0.005 °C
Temperature coefficient	0.002 °C °C^{-1}
Supply	4.5 V to 5.5 V (nom. 5 V), 1.3 A

Table F.4: Key specifications of the Koheron TEC100L-50 temperature controller.

F.4 Balanced Photodetector (Koheron PD100B-AC)

The PD100B is an amplified balanced detector with two mounted InGaAs photodiodes feeding a differential transimpedance amplifier. The AC-coupled version was used, so the DC photocurrent from the LO is blocked above 160 Hz.

Parameter	**PD100B-AC**
Photodiodes	InGaAs, 300 µm active diameter, FC connectors
Wavelength range	900 nm to 1700 nm
Peak responsivity	$0.9\,\mathrm{A\,W^{-1}}$
Optical input power (per diode)	0 mW to 1.5 mW
Transimpedance gain	$39\,\mathrm{kV\,A^{-1}}$
Small-signal bandwidth (3 dB)	160 Hz to 100×10^6 Hz
Input current noise density (10 MHz)	$8\,\mathrm{pA}/\sqrt{\mathrm{Hz}}$
CMRR (1 MHz)	35 dB
Output	±3 V range, 50 Ω, SMA
Supplies	±6 V nominal (5.5 V to 12 V)

Table F.5: Key specifications of the Koheron PD100B-AC balanced photodetector.

Appendix G

Supplementary Results: Experiment A

G.1 A2: 5–20 MHz Band-averaged PSD per Laser Current

Table G.1 gives the data plotted in Figure 4.3: the detector output PSD averaged over the 5–20 MHz band for each laser drive current with balanced detection. The laser-off (electronic) floor is $8.061 \times 10^{-14}\,\mathrm{V^2\,Hz^{-1}}$; the excess above this floor is taken as the shot noise S, and the last column gives the total noise T relative to the shot-noise limit, $10\log_{10}(T/S)$, which performance requirement P1 requires to be ≤ 1 dB.

Current (mA)	Total PSD T (10^{-13} V^2/Hz)	Excess $S = T - E$ (10^{-13} V^2/Hz)	Voltage noise (nV/$\sqrt{\mathrm{Hz}}$)	T/E (dB)	T/S (dB)
7	1.1127	0.3066	333.6	1.40	5.60
11	1.3797	0.5736	371.4	2.33	3.81
15	1.8440	1.0379	429.4	3.59	2.50
23	2.8416	2.0355	533.1	5.47	1.45
31	3.6844	2.8783	607.0	6.60	1.07
38	4.6072	3.8011	678.8	7.57	0.84
48	5.2954	4.4893	727.7	8.18	0.72

Table G.1: A2: band-averaged (5–20 MHz) detector PSD with balanced detection versus laser drive current. $E = 8.061 \times 10^{-14}\,\mathrm{V^2\,Hz^{-1}}$ is the laser-off electronic floor.

G.2 A3: Supplementary Experiment on RIN Cancellation

Figure G.1 gives the 5 - 20MHz averaged PSD of the detected noise, when only a single photodetector input of the PD100B is used - effectively turning it into a singe-ended detector. The electronic noise floor will remain the same, but the total noise PSD averages will now also contain RIN in addition to shot-noise and electronic noise. Before analysis of this plot can commence, it is important to note that in the case of balanced detection, the shot-noise of each detector is uncorrelated, and will therefore add together at the output. Thus, in the single-ended case, the shot-noise at the output for a given laser current is expected to be half that of the balanced detection case. Any excess noise above this half-power shot-noise level is thus contributed from the electronic noise, and the uncancelled RIN.

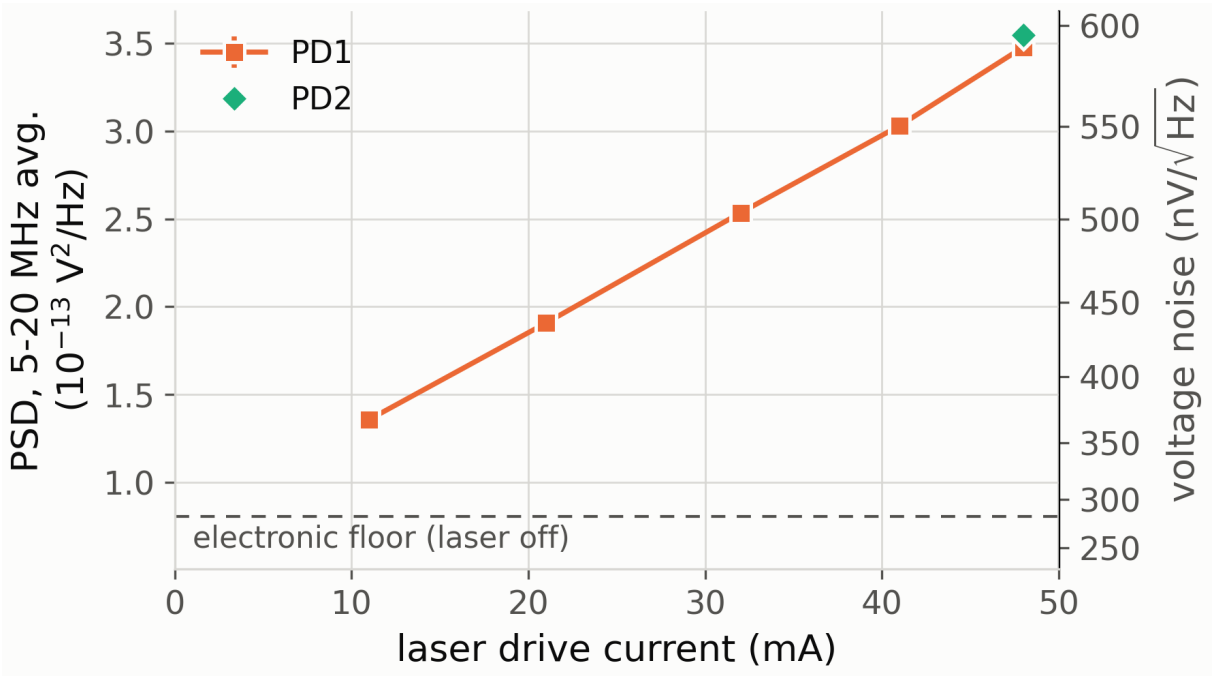


Figure G.1: A3: 5 - 20MHz Average PSD vs Laser Current with Single-Ended Detection

It can be seen in Figure G.1 that indeed, the total noise is lower than in Figure 4.3, as expected. Observing the total noise at the operating point of 48mA however, it can be seen that a value of approximately $3.5 \times 10^{-13} V^2/Hz$ is obtained, which gives $2.694 \times 10^{-13} V^2/Hz$ once electronic noise has been subtracted. In the balanced detection case at the same laser current, a value of $5.2954 \times 10^{-13} V^2/Hz$ was obtained - subtracting electronic noise and halving this gives $2.245 \times 10^{-13} V^2/Hz$. This clearly indicates that the noise power in the single-ended case is higher than simply half of the shot-noise in

the balanced detection case, thus proving that indeed there is excess RIN present. This excess RIN noise can be calculated:

$$RIN_{excess} = (2.694 - 2.245) \times 10^{-13} = 0.449 \times 10^{-13} V^2/Hz \tag{G.1}$$

Converting to dB:

$$10log_{10}(1 + \frac{RIN_{excess}}{S}) = 10log_{10}(1 + \frac{0.449 \times 10^{-13}}{2.245 \times 10^{-13}}) = 0.79dB \tag{G.2}$$

Thus, RIN contributes an additional 0.79dB of noise above the ideal shot-noise limit at the 48mA operating point. Balanced detection allows this excess noise to be removed, as illustrated in the results of A2 - allowing for a better SNR.

Appendix H

Supplementary Results: Experiment D

H.1 Raw PSDs per Wind-tunnel Setpoint

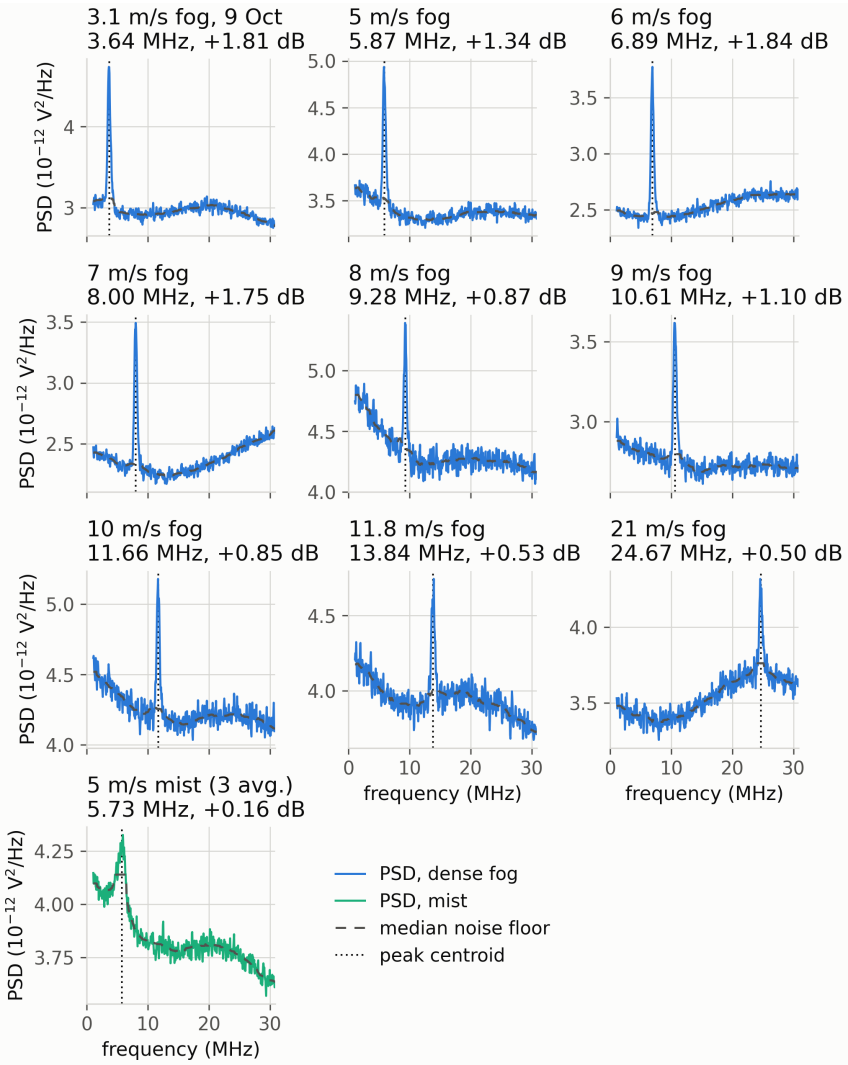


Figure H.1: Grid of PSD plots for each wind-tunnel setpoint

H.2 Raw Noise and Peak Statistics

Table H.1 and Table H.2 give the un-normalised counterparts of Table 5.2 and Table 5.3: the same Doppler peaks and noise bands, but with the statistics taken on the raw PSD rather than on the PSD divided by its running-median floor. Because the raw floor slopes across the noise band, its standard deviation here includes that slope as well as the random fluctuation, which makes the raw mean $+5\sigma$ threshold the more conservative of the two.

Setpoint	f_D	PSD ($10^{-12}\,\mathrm{V^2/Hz}$)				Peak/mean	Excess
(m/s)	(MHz)	floor mean	floor σ	mean $+5\sigma$	at f_D	(dB)	($10^{-7}\,\mathrm{V^2}$)
3.1	3.637	3.01	0.078	3.41	4.73	1.96	10.25
5.0	5.868	3.45	0.075	3.82	4.79	1.43	8.87
6.0	6.892	2.44	0.031	2.59	3.78	1.90	7.55
7.0	7.997	2.30	0.040	2.50	3.49	1.82	7.04
8.0	9.280	4.30	0.082	4.71	5.36	0.95	5.99
9.0	10.611	2.74	0.042	2.95	3.60	1.19	6.08
10.0	11.664	4.20	0.054	4.46	5.18	0.92	6.00
11.8	13.837	3.94	0.053	4.20	4.51	0.59	5.27
21.0	24.674	3.69	0.052	3.95	4.22	0.59	4.01

Table H.1: D: raw (un-normalised) noise and Doppler-peak statistics for dense fog-seeded air

Setpoint	Captures	f_D	PSD ($10^{-12}\,\mathrm{V^2/Hz}$)				Peak/mean	Excess
(m/s)	averaged	(MHz)	floor mean	floor σ	mean $+5\sigma$	at f_D	(dB)	($10^{-7}\,\mathrm{V^2}$)
5.0	1	5.730	4.48	0.090	4.93	4.80†	0.30	4.38
5.0	2	5.716	3.83	0.087	4.27	4.08†	0.28	4.45
5.0	3	5.726	4.00	0.081	4.41	4.29†	0.30	3.75

† PSD at f_D below the raw mean $+5\sigma$ threshold. The detection verdicts in Table 5.3 use the floor-normalised PSD instead, on which the 5 m/s peak clears its threshold with three captures.

Table H.2: D: raw (un-normalised) noise and Doppler-peak statistics for light mist-seeded air.

H.3 Velocity Accuracy: Peak Centroid vs Setpoint

Table H.3 gives the data behind Figure 5.7: for each setpoint, the Doppler-peak centroid, the line-of-sight velocity it implies, the velocity expected from the setpoint and the 35°

beam angle, and the error between the two.

Setpoint U (m/s)	f_D (MHz)	Measured v_{LOS} $= f_D\lambda/2$ (m/s)	Expected v_{LOS} $= U\cos 35^\circ$ (m/s)	Error (%)
Collimator facing downstream (original rig)				
3.1	3.637	2.82	2.54	+11.0
5.0	5.868	4.55	4.10	+11.0
6.0	6.892	5.34	4.91	+8.7
7.0	7.997	6.20	5.73	+8.1
8.0	9.280	7.19	6.55	+9.8
9.0	10.611	8.22	7.37	+11.6
10.0	11.664	9.04	8.19	+10.4
11.8	13.837	10.72	9.67	+10.9
21.0	24.674	19.12	17.20	+11.2
Fit through origin: 1.170(3) MHz per m/s, +10.7 % above the expected 1.057 MHz per m/s				
Collimator facing upstream (rotated rig)				
3.9	4.269	3.31	3.19	+3.6
5.3	5.733	4.44	4.34	+2.3
6.1	6.802	5.27	5.00	+5.5
6.3	6.816	5.28	5.16	+2.4
7.5	8.422	6.53	6.14	+6.2
8.0	9.105	7.06	6.55	+7.7
9.0	10.265	7.95	7.37	+7.9
10.0	11.337	8.79	8.19	+7.3
21.0	23.864	18.49	17.20	+7.5
Fit through origin: 1.130(6) MHz per m/s, +6.9 % above the expected 1.057 MHz per m/s				

Table H.3: D: Doppler-peak centroid and line-of-sight velocity against the wind-tunnel setpoint for both rig orientations, in dense fog-seeded air

Appendix I

Declaration on use of AI tools

I.1 Report writing

The contents of this report are my own work. Artificial intelligence use was limited to formatting of tables, figures and plots, and was used to proofread contents for possible errors and improvements in clarity and conciseness. Example prompts for these use cases are given below. Note that this is not exhaustive, but serves as an example for some of the use cases.

I.1.1 Formatting of tables, figures and plots

Prompt

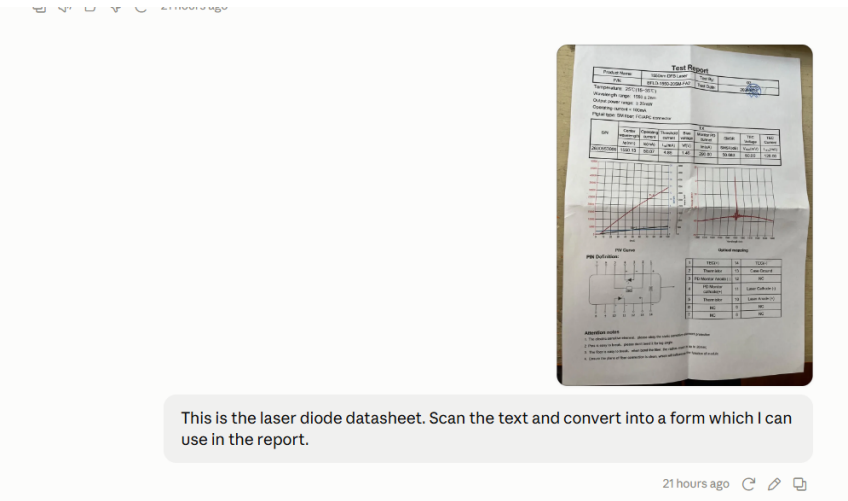


Figure I.1: Example prompt to format tables and figures for the laser datasheet appendix

Response

Here's the test report transcribed as two tables plus a note on the curves, ready to drop into the `appendix:laser` section of `appendix_E_datasheets.tex`. The two plots can't be transcribed, so I've suggested keeping a cropped photo of them.

```
\section{DFB Laser Diode (BFLD-1550-20SM-FA2)}
\label{appendix:laser}
The laser diode was supplied with the manufacturer's test report reproduced below (
serial number 2630553009, tested 21 August 2026). General specifications: operating temperature \
SI{25}{\celsius} (15--\SI{35}{\celsius}); wavelength range $1550 \pm 2$\,nm; output power $\geq \
SI{20}{\milli\watt}$; operating current $< \SI{100}{\milli\ampere}$; single-mode fibre pigtail
with FC/APC connector; 14-pin butterfly package.

\begin{table}[H]
    \centering
    \begin{tabular}{@{}llr@{}}
        \toprule
        \textbf{Parameter} & \textbf{Symbol} & \textbf{Measured value} \\
        \midrule
        Centre wavelength            & $\lambda_p$          & \SI{1550.13}{\
nano\metre} \\
        Operating current (at \SI{20}{\milli\watt}) & $I_o$ & \SI{58.07}{\
milli\ampere} \\
        Threshold current            & $I_{\mathrm{TH}}$    & \SI{4.86}{\milli
\ampere} \\
        Bias (forward) voltage       & $V_f$                & \SI{1.46}{\volt}
 \\
        Monitor photodiode current   & $I_m$                & \SI{290}{\micro\
ampere} \\
        Side-mode suppression ratio  & SMSR                 & \SI{50.68}{\
decibel} \\
        TEC voltage                  & $V_{\mathrm{TEC}}$   & \SI{50}{\milli\
volt} \\
        TEC current                  & $I_{\mathrm{TEC}}$   & \SI{120}{\milli\
ampere} \\
        \bottomrule
    \end{tabular}
    \caption{Manufacturer test-report values for the BFLD-1550-20SM-FA2 laser
diode (S/N 2630553009) at \SI{25}{\celsius}.}
    \label{tab:laser-testreport}
\end{table}

\begin{table}[H]
    \centering
    \begin{tabular}{@{}cl@{\qquad}cl@{}}
        \toprule
        \textbf{Pin} & \textbf{Function} & \textbf{Pin} & \textbf{Function} \\
        \midrule
        1 & TEC ($+$)                  & 14 & TEC ($-$) \\
        2 & Thermistor                 & 13 & Case ground \\
        3 & PD monitor anode ($-$)     & 12 & NC \\
        4 & PD monitor cathode ($+$)   & 11 & Laser cathode ($-$) \\
        5 & Thermistor                 & 10 & Laser anode ($+$) \\
        6 & NC                         &  9 & NC \\
        7 & NC                         &  8 & NC \\
        \bottomrule
```

```
                \end{tabular}
                \caption{Pin definition of the 14-pin butterfly package (thermistor \SI{10}{\
kilo\ohm}).}
                \label{tab:laser-pins}
        \end{table}

        The test report also gives the measured P--I--V curve and the optical spectrum (\SI
{1500}{}--\SI{1600}{\nano\metre}), reproduced in \autoref{fig:laser-testreport-curves}. The P--I
curve is linear above threshold at approximately \SI{0.35}{\milli\watt\per\milli\ampere}, giving
roughly \SI{17}{\milli\watt} at the \SI{48}{\milli\ampere} operating point used in this project (\
autoref{sec:testingA}).

        \begin{figure}[H]
                \centering
                % TODO: crop the two plots (PIV curve and optical mapping) from the photo of
                % the test report and save as laser_testreport_curves.png
                \includegraphics[width=0.9\linewidth]{laser_testreport_curves.png}
                \caption{Manufacturer P--I--V curve (left) and optical spectrum (right) for
the supplied diode, from the test report.}
                \label{fig:laser-testreport-curves}
        \end{figure}
```

This response was then checked against the original datasheet and added as part of the appendix.

I.1.2 Improvements in clarity and conciseness

Prompt

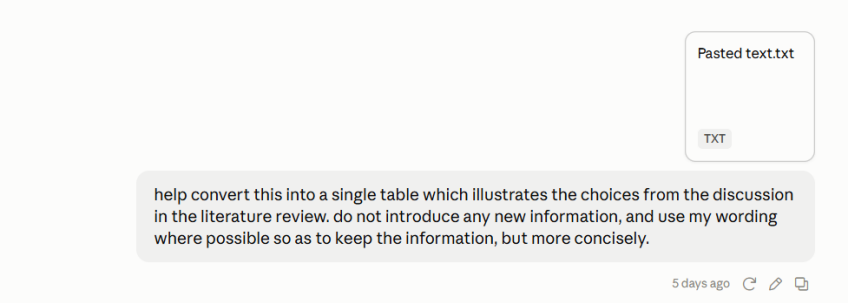


Figure I.2: Example prompt to improve clarity and conciseness of architecture decisions

Response

Here's the section as a single table in LaTeX, reusing your wording and citations throughout. It needs `tabularx` and `booktabs` in the preamble.

```
            % Preamble: \usepackage{tabularx,booktabs}
            % Optional: \newcolumntype{Y}{>{\raggedright\arraybackslash}X}  (stops wide word gaps
in narrow columns)

            \section{System Architecture}
            \label{sec:design-sysArch}
            From the reviewed literature in \autoref{sec:litReview}, it was decided that an \
textbf{all-fibre, coherent detection, continuous-wave, monostatic, homodyne} DWL system with \
textbf{balanced detection} would be built, with sampling and digitisation handled by a benchtop
oscilloscope, and post-processing done on a laptop computer. \autoref{tab:arch} summarises why
each architectural choice was made with respect to the project goals, and the tradeoffs accepted.

            \begin{table}[htbp]
                    \centering
                    \small
                    \renewcommand{\arraystretch}{1.25}
                    \begin{tabularx}{\textwidth}{@{}>{\raggedright\arraybackslash}p{2.1cm} >{\
raggedright\arraybackslash}p{2.2cm} Y Y@{}}
                            \toprule
                            \textbf{Choice} & \textbf{Alternative} & \textbf{Reason for choice} &
\textbf{Accepted tradeoff} \\
                            \midrule
                            Coherent detection & Direct detection &
                            Direct detection requires the Rayleigh/Mie scattering ratio and
temperature, an absolutely stabilised laser, etalon control to 1\,mK, and very precise alignment
and calibration \cite{korb_theory_1998,colberg_diode-laser-based_2026,werner_doppler_2005}, which
are substantial challenges to an undergraduate project. Coherent detection is far more common in
the literature, permits shot-noise-limited operation, and rejects unwanted light not mode-matched
to the LO (e.g.\ other LiDAR systems, or the sun) \cite{hill_remote_nodate}. &
                            Cannot use molecular backscatter when aerosol backscatter is not
available; irrelevant in an aerosol-rich, controlled lab environment. \\
                            \addlinespace
                            Homodyne & Heterodyne (AOM in LO or transmit arm) \cite{
harris_role_1998,abdelazim_development_2015} &
                            Places the Doppler shift at baseband with no additional components; an
 AOM adds needless complexity and cost to a budget-constrained project. &
                            Only the velocity \emph{magnitude} is measured (\autoref{sec:lit-
homohetero}); acceptable as the flow direction will always be known. An AOM can be added in a
future iteration without redesign. \\
                            \addlinespace
                            Balanced detection & Single-ended detection &
                            3\,dB SNR improvement in shot-noise-limited operation (\autoref{sec:
lit-BalancedDetection}), and RIN is largely cancelled, allowing the system to better approach the
shot-noise limit (Appendix~\ref{appendix:RIN}). &
                            -- \\
                            \addlinespace
                            Monostatic optics & Bistatic optics &
                            A single aperture serves both transmit and receive, separated by a
fibre circulator; avoids the greatly increased alignment complexity, and cost of a second set of
optics and alignment equipment \cite{cariou2006laser,harris_role_1998,bennett_doppler_2003}. &
                            Back-reflections from the transmit path reach the detector and raise
the noise floor through PIIN (\autoref{sec:lit-linewidth}). \\
```

```
                            \addlinespace
                            Continuous wave & Pulsed &
                            Needs no pulse or receive timing, which simplifies design and reduces
costs, and is more effective at the short ranges (tens of cm to 1--2\,m) of a benchtop system, as
pulsed systems are ineffective below tens of metres \cite{hill_remote_nodate}. &
                            Reduced spatial resolution at long ranges, as resolution is defined by
 focus rather than gating (\autoref{sec:lit-pulsedCW}); irrelevant at benchtop range. \\
                            \addlinespace
                            All-fibre, non-PM & Free-space; PM fibre &
                            Single-mode fibre makes LO--return mode-matching automatic, removes
free-space alignment complexities, and gives access to widely and cheaply available telecom
components \cite{bennett_doppler_2003,abari_all-fiber_2014,abdelazim_development_2015}. Non-PM
fibre is used owing to cost. &
                            LO and return polarisation is not fixed, reducing the heterodyne
efficiency and the measurable Doppler peak \cite{cariou2006laser}. \\
                            \bottomrule
                    \end{tabularx}
                    \caption{Architectural choices, with reasons and accepted tradeoffs, as
synthesised from the literature in \autoref{sec:litReview}. PM: polarisation-maintaining.}
                    \label{tab:arch}
            \end{table}
```

A few notes:

- **Homodyne in the opening sentence:** I added it to the bolded list. That isn't new information, since the section already concluded homodyne, but it was missing from the original list.
- **Balanced detection trade-off:** your text didn't give one, so the cell is a dash rather than something I made up.
- **Removed subsection labels:** the "Coherent detection" subsection reused `sec:design-sysArch`, which clashed with the section's own label. It's gone now. If anything else in the report referenced the deleted subsections, for example the `??` in §3.4.5 pointing to CW, change those references to `\autoref{tab:arch}`.
- **Length:** this should come out at about two-thirds of a page, compared with roughly 2.5 pages before.

Initially, Table 3.1 was written as prose over 3 - 4 pages. This response was then used so as to reduce it to a single page table, maintaining my own wording used in the section originally. The table was checked against the writing originally used, and was then inserted into the report.

I.2 Conceptual understanding

An artificial intelligence model was used to assist in conceptual understanding of design considerations and results of the testing. The model was given access to all of the research papers in the literature review, as well as a document detailing my testing plans and work so far. All responses were checked against the relevant sources which they cite for correctness. An example of a prompt used for this, and the response is given below.

I.2.1 Conceptual understanding of PIIN

Prompt

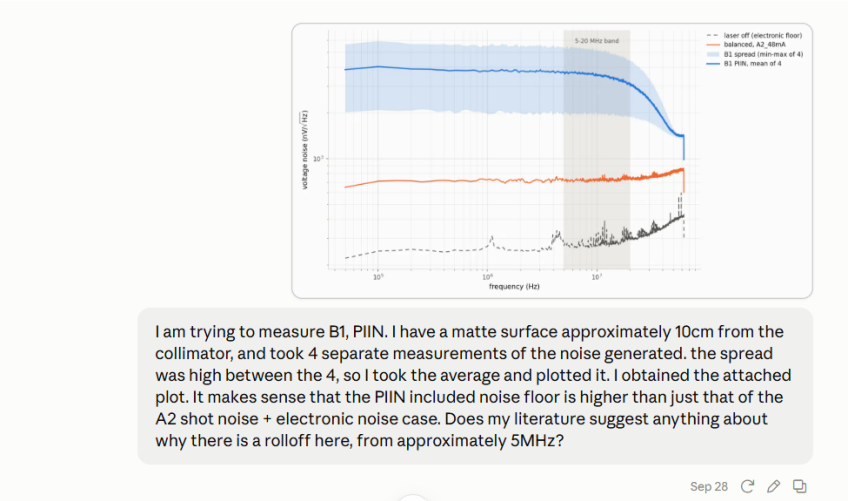


Figure I.3: Example prompt used to assist in conceptual understanding of PIIN and PIIN roll-off

Response

Let me check what Harris et al. and Jiang et al. say about the spectral shape before answering.
[Searches the Harris et al. (1998) paper for its discussion of delay, coherence and spurious reflections, and reads Section 5.]
Yes. Harris et al. (1998), your coherence-length paper, covers this almost exactly in its Section 5, "Excess local oscillator noise". Jiang et al. (2019) supports it. The roll-off doesn't come from the laser linewidth. It comes from the **delay between the LO and the reflected light**, and its frequency tells you how long that delay is.

What Harris says

- In a monostatic fibre lidar, weak reflections "from within the coupler and from fibre connectors" interfere with the LO. Because they travel different distances, "this gives rise to delayed self-heterodyne spectra", and the noise floor "has essentially the same origin as the incoherent component in the delayed self-heterodyne spectra" of their Section 3.
- The broadband part of that spectrum has "ripples… separated by the inverse of the delay time", and "the relative distances can be inferred from the shape of the spectra". In other words, the shape of the spectrum encodes the path mismatch.
- For delays much shorter than the coherence length, the low-frequency excess noise scales with the **square of the fibre-path mismatch** and inversely with the coherence length (their Eqs. 3–5). Adding 4 m of fibre made the excess noise rise "dramatically".
- They saw excess LO noise 5–30 dB above shot noise that was "extremely unstable and very sensitive to small disturbances of the optical fibres". That describes your 4-capture spread.

Jiang et al. add the observation that PIIN is strongest near zero frequency and can sit orders of magnitude above shot noise. That is qualitatively what your plot shows: a flat region at low frequency that is well above the A2 floor.

The mechanism, and why the roll-off appears where it does

The reflection interferes with the LO at a slowly drifting static phase φ_0. The laser's phase noise over the delay τ, $\Delta\varphi(t) = \varphi(t) - \varphi(t-\tau)$, turns into intensity noise. For a laser with a Lorentzian linewidth $\Delta\nu$ and τ much shorter than the coherence time:

$$S_{\mathrm{PIIN}}(f) \propto P_{\mathrm{LO}}\, P_r\, \sin^2\varphi_0\, \Delta\nu\, \frac{\sin^2(\pi f\tau)}{f^2} = S_{\mathrm{PIIN}}(0)\,\mathrm{sinc}^2(f\tau), \qquad S_{\mathrm{PIIN}}(0) \propto \Delta\nu\,\tau^2$$

- It is **flat** at low frequencies, starts to fall above about $1/(4\tau)$, is **3 dB down at about** $0.44/\tau$, and has its first null at $\mathbf{1}/\tau$. The nulls are Harris's "ripples".
- The $\Delta\nu\,\tau^2$ factor is Harris's "$\propto$ (fibre length)^2 / coherence length".
- It is **not** a Lorentzian with a 2 MHz width. That shape only applies when τ is much longer than the coherence time, and here $\tau_c = 1/(\pi\Delta\nu) \approx 160$ ns, or about 33 m of fibre. Your flat region extends well past 2 MHz, which confirms you're in the short-delay regime Harris describes.

What your curve implies

I read your plot approximately. I removed the A2 floor in power and normalised by the A2 shot-noise excess, which is white at the diode, so dividing by it cancels the detector's gain

peaking and the scope's roll-off. The PIIN then falls by about $-2.3\,\mathrm{dB}$ at $20\,\mathrm{MHz}$, $-6.6\,\mathrm{dB}$ at $40\,\mathrm{MHz}$ and $-12\,\mathrm{dB}$ at $60\,\mathrm{MHz}$. A sinc^2 fit to those points gives $\boldsymbol{\tau \approx 13}$**–**$\boldsymbol{20\,\mathrm{ns}}$, which is a path mismatch of about **3–4 m of fibre** (at $0.204\,\mathrm{m}$ per ns). That's plausible given the circulator pigtails, the collimator patch cord and the coupler leads. The spread in the fitted τ across frequency suggests that several reflections with different delays are contributing, which Harris also describes.
This gives you a good check for the report. Add up the fibre lengths from the splitter to the coupler along the LO path, and along the path through the circulator to the collimator and back. Their difference should give τ, and its sinc^2 prediction should match the measured roll-off. Worth remembering: the $10\,\mathrm{cm}$ of free space to your matte surface adds only about $0.7\,\mathrm{ns}$, so the delay is dominated by fibre.

Why the 4 captures differ so much

The $\sin^2\varphi_0$ term explains it. The interferometric phase between the LO and the reflection wanders with temperature and with fibre movement. Near quadrature ($\varphi_0 = 90°$) PIIN is at its maximum; near $\varphi_0 = 0$ it almost vanishes. Your min–max band is about $2\times$ in voltage ($6\,\mathrm{dB}$ in power) at low frequency and keeps the same shape across frequency. That is what a phase factor would do: change the amplitude and leave the shape alone. (Harris's lidar used an $80\,\mathrm{MHz}$ frequency offset, which averages this phase out; your homodyne system doesn't, which is why yours fluctuates.)
For the analysis:

- **Four captures are too few** to average over a random $\sin^2\varphi_0$. Take 20–50 captures spread over several minutes. Report the **mean** (which tends to half the maximum) as typical performance, and the **maximum** as the worst case (quadrature).
- **Fit the shape rather than the level.** Normalise each capture by its own low-frequency value; the curves should collapse onto one sinc^2, and τ is the robust result.
- **Check which reflection dominates** with the B0–B3 breakdown from before. Note that $S_{\mathrm{PIIN}}(0)$ scales with τ^2, so a reflection at the end of a long fibre path matters far more than an equally strong one close to the coupler.

I.3 Data acquisition and processing scripts

An artificial intelligence model was used for assistance in writing Python scripts for data acquisition and subsequent processing of that data. Examples of prompts given and responses to those prompts are given below.

I.3.1 Plotting of PIIN results

Prompt

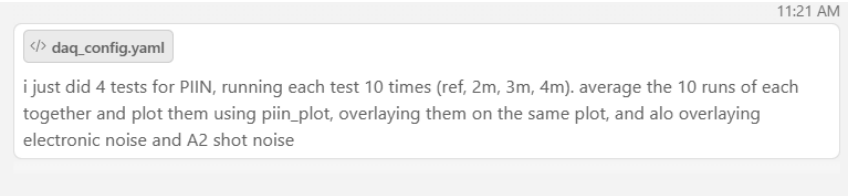


Figure I.4: Example prompt given to generate plots for PIIN test results from raw data

Response

[Lists the project files and the most recent captures, and searches for the plotting script.]
[Reads ***psd_piin.py*** *and the metadata of one PIIN run, then checks the A1 and A2 capture settings and the loader and Welch code.]*
[Loads its plotting guidelines and reads the reference colour palette.]
Checking the run data for outliers, then extending `psd_piin.py` to plot several series.
[Computes the 5–20 MHz band level, clipping, interruption flag, scope scale and laser current for each of the 40 runs.]
All 40 runs are clean (no clipping, none interrupted, all at 48 mA). Now generalising `psd_piin.py` to take several series.
[Checks that ***psd_piin.py*** *has no uncommitted changes, then rewrites it to accept several series.]*
[Runs the original single-series plot as a regression check, and the new overlay plot. Both fail on a file with no raw records.]
The data folder now has non-capture `.npz` files (the `*_stack.npz`) that the loader trips on — a pre-existing issue. I'll filter on the filename tag before loading, which also skips loading hundreds of unrelated captures.
[Edits the file selection, reruns both plots, and views the resulting figure.]
I averaged each set of 10 runs and plotted all four on one figure, with the laser-off electronic floor (A1) and the A2 shot noise (balanced, 48 mA) underneath: `figures/psd_piin_lengths.png`.
[Embeds the figure.]
I found `psd_piin.py` (the "piin_plot" script); it only handled one series, so I extended it rather than writing a new script. Run this to regenerate the figure:

```
python psd_piin.py --series PIIN_ref PIIN_2m PIIN_3m PIIN_4m --label ref 2m 3m 4m --ref-label "A2 shot noise (balanced, 48 mA)" --out figures/psd_piin_lengths.png
```

Changes to `psd_piin.py`:

- **Several series:** `--series` and `--label` now take one entry per series. Each series gets its own mean curve, a light min-max band showing the spread between runs (`--no-spread` turns the bands off), and its own printed statistics.
- `--ref-label`**:** sets the legend text for the A2 curve.
- **Crash fix:** the script crashed on the `*_stack.npz` files now in `data/`, because they hold no raw records. It now picks files by the tag in the filename before loading them, which also means it doesn't load every unrelated capture. The original `python psd_piin.py` still runs; its numbers are below and the figure is in the scratchpad.

Data checks: all 40 runs are clean, with no clipping, none interrupted, and all at 48 mA. Two filenames are mislabelled: `PIIN_ref6` appears twice with no `ref7`, and `PIIN_4m_1` appears twice with no `4m_2`. Each set still has 10 runs, and all were included.

Results in the 5–20 MHz band:

Series	Amplitude (nV/$\sqrt{\text{Hz}}$)	Power vs A2	Spread between runs (1 SD)
A1 floor	284	—	—
A2 shot noise	728	1×	—
ref	9247	161×	±30%
2m	5279	53×	±14%
3m	4288	35×	±17%
4m	1407	3.7×	±28%

Two things stand out:

- **4m:** it is far lower than 3m, close to the shot-noise level. That's a much bigger step than 2m→3m.
- **ref:** it is the only curve that falls off sharply above about 25 MHz, down to around the 2m/3m level near 45 MHz.

Both may come from the setup rather than the analysis, so they're worth checking before you write them up. Nothing is committed.

The plots generated here were then checked against the raw data obtained and were used in the report.

I.3.2 Windtunnel data acquisition script

Prompt

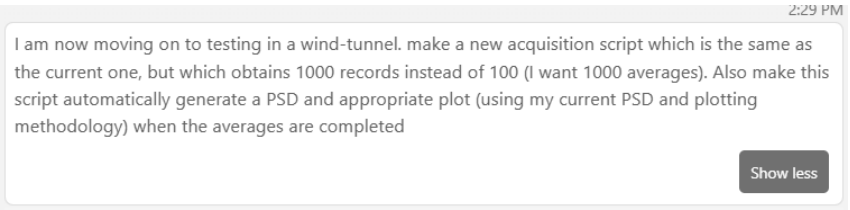


Figure I.5: Example prompt for generating an acquisition script for acquiring oscilloscope data for wind-tunnel testing

Response

[Reads the acquisition script and its configuration file, and checks the PSD output format of `run_welch.py`*.]*
[Writes `acquire_windtunnel.py`*.]*
I'll swap that hand-rolled `--config` lookup for a small parser, then test it.
[Edits the argument handling, then tests the PSD-only path on an existing capture with a reference.]
[Removes the test outputs and runs the full flow with a simulated scope.]
The test caught a real bug. argparse treats `--n` as an abbreviation of `--nperseg`/`--no-psd`, so `--n 50` would never reach `Acquisition.py`. The fix is to turn off abbreviation matching.
[Disables abbreviation matching and reruns the simulated full flow.]
[Changes the script to save only the averaged PSD, then rechecks the file size and the `--dry-run` *path.]*
The PSD file is now 63 kB instead of 47 MB, and `--dry-run` correctly saves nothing and skips the PSD. Checking the plot:
[Views the test plot.]
I've written `acquire_windtunnel.py`, which captures 1000 records and then automatically makes the PSD and plot. I couldn't test it on the scope itself; I ran it end to end with a simulated scope instead.

```
python acquire_windtunnel.py --tag WT_noflow
python acquire_windtunnel.py --tag WT_10mps --ref WT_noflow --note "fan 40
%"
```

How it works

- **Capture:** it calls `Acquisition.py` unchanged, so the scope setup, `daq_config.yaml`, the `.npz`/`.json` files and the progress output are all the same. The only difference is that the record count defaults to 1000. All `Acquisition.py` options still work (`--tag`,

`--note`, `--dry-run`, `--no-configure`, `--list`), and `--n` still overrides the 1000. Ctrl-C still stops early and saves; the PSD is then computed from whatever records were captured.

- **PSD:** it uses the same method as `psd_rpm.py`: Welch with a Hann window and 50 % overlap, each record transformed separately, averaged in V^2/Hz, with `nperseg` 12500 (10 kHz bins). 1000 records give 3000 averages, so the curve scatters about 0.08 dB per bin, against 0.24 dB at 100 records.
- **Plot:** it's styled like the per-RPM plots: log-log in $nV/\sqrt{Hz}$, with a dashed reference line if you pass `--ref` (it uses the most recent capture with that tag). It's saved next to the capture as `<stem>_psd.png`, along with `<stem>_psd.npz` holding the frequencies, the PSD, the sample rate and the settings used.
- **Re-plotting:** if the PSD step ever fails, the capture is still saved. You can redo the PSD on any file with `python acquire_windtunnel.py --psd data/<file>.npz --ref WT_noflow`.

Differences from `run_welch.py`

- **Chunked PSD:** it's computed 100 records at a time, because all 1000 at once would need about 1 GB of RAM. The result is identical.
- **No per-record PSDs saved:** at 1000 records they made the file 47 MB, bigger than the raw capture. Without them it's 63 kB, and they can be recomputed from the raw file.

Test results with the simulated scope

- The full run (capture, save, PSD, reference overlay, plot) worked. A simulated 20 mV rms of noise plus a 4 mV tone came out at the expected total.
- `--dry-run` saves nothing and skips the PSD, as before.
- I fixed one bug the test found: `--n 50` was being read as the new script's `--nperseg` instead of reaching `Acquisition.py`.
- One test run wrote two files into your real `data/` folder; I deleted them straight away. The other test files are only in the scratchpad.

Each run will take about 100 s and write about 42 MB of raw data.
Check the Doppler frequency range before you start: at 1550 nm the shift is 1.29 MHz per m/s along the beam. In the RPM data the detector starts rolling off around 20 MHz, so flows above about 15 m/s along the beam will fall into the roll-off. A beam angle to the flow lowers the shift.

The script generated by the response was then modified, e.g. to only use 500 records

rather than 1000, and to use nperseg = 2500 instead of 12 500. The script's functionality was verified by testing the acquisition of a ground-truth signal from a function generator and verifying the correct output, and was thereafter used in data acquisition in the wind-tunnel testing phase.

Appendix J

Code and Additional Materials

All of the code used for data acquisition, processing and plotting can be found at the GitHub repository for this project (https://github.com/thomasbkruger/Final-Year-Project).

Appendix K

Ethics Forms



PRE-SCREENING QUESTIONNAIRE OUTCOME LETTER

STU-EBE-2026-PSQ002612

2026/08/11

Dear Thomas Kruger,

Your Ethics pre-screening questionnaire (PSQ) has been evaluated by your departmental ethics representative. Based on the information supplied in your PSQ, it has been determined that you do not need to make a full ethics application for the research project in question.

You may proceed with your research project titled:

LiDAR-Based Wind Speed Measurement

Please note that should aspect(s) of your current project change, you should submit a new PSQ in order to determine whether the changed aspects increase the ethical risks of your project. It may be the case that project changes could require a full ethics application and review process.

Regards,

Faculty Research Ethics Committee

Figure K.1